

CCPS 731 Software Engineering I

Ryerson University

Term Project Report: Gmail Template Engine

Dr. Muhammad Naeem Irfan

Alex Ladd 500808377

Hemanshu Bhargav 500588413

Yashodhar Dave 500308864

Project Proposal (SUB 1)

Introduction

Many operations' (marketing, sales, production, supply chain, IT, customer service, HR etc.) staff often sends confidential messages— in a sense, they are for intended recipients only. However, many of these are common in content, at least through the perspective of users sending them on a daily basis. Our tool seeks to not automate the task of emailing, but rather empower the user to create their own email template engine.

Objectives

We intend to create a customizable Gmail add-on which allows email templating and personalized message delivery. The email add-on will host a light-weight database (the size of a small table) which will be populated based on the custom user profiles. The user will have the freedom to design prefilled messages for as many scenarios as they require, with one-click execution to automatically fill the selected/current email message with the desired template.

Target Customers

Working professionals in dire need of time in their busy lifestyles. This tool can be used by anyone who uses email as a significant communication channel in their daily life. Our focus will cater on those who send frequent repetitive emails to individuals, emails which cannot be automated as the recipients and content differs slightly each time, but nonetheless is repetitive for the user.

Value Proposition

The human aspect of the interface remains critical and relies on the user's domain knowledge to customize their profile with their unique needs—rather than simply relying on generic profiles force fed to the consumer. The user is given the power to save their own custom messages for specific scenarios without any need to write a line of code. Furthermore, the user has all of these messages available from the touch of a button. A simple click will allow the user to automatically fill their selected message with the desired custom message— so that it can be sent or modified as the user sees fit.

Application Features and Description

1st Iteration

- Allow users to create a profile and save custom email templates for various scenarios
- Extract/read subject-line of current/selected email
- Use pre-populated database to auto-generate emails
- Bonus: Move most frequently used templates under “recent section”

2nd Iteration

- List Google Calendar events for possible automated actions
- Log user behaviour (based on previous templates used, suggest most frequently used templates)

- Bonus: Use machine learning algorithms or available Google intelligence to suggest templates based on past responses (regardless of frequency of use)

Tools and Resources

Chrome Web Store

Gmail Addon

NotePad++

Google Drive REST API (for source control)

Google Documentation

Javascript JQuery Framework

Challenge

1st iteration

Our biggest challenge will be creating a light-weight add-on given the sheer complexity of our task. Since our add-on relies on a database, we will have to devise a way for the database to link to the user's Google account and either be hosted on Google's servers or on the user's local device. The desired method is to utilize the user's Google account so that the user's profile is available anytime and anywhere along with the rest of their Google services. Currently, no database server support exists for Google add-ons, so we will be using Google Spreadsheets as our database. This brings with it its own challenges as we will be limited to only certain operations and not have the full utilities of modern database applications. We assume that the size of the spreadsheet will not grow too large, in a manner that may negatively impact the performance/speed of the add-on.

As the users clicks the add-on, an interface appears with the user's message options. Our add-on script should be able to retrieve:

- user's inbox messages
- user's sent messages
- templates stored in the database—which must be available based on input
- mechanism for lexical analysis

Sample scenarios user's may use as templates:

- stock/inventory updates
- medical test updates
- waste analysis
- lendings/owings
- username/password reset
- product invitations

Software Process Model (SUB 2)

Our team decided on the “Agile Model” for the development of our product, mainly due to our team’s size and preference on an iterative approach. Given the lack of customer availability in our development (particularly as this project has not been announced to the public, nor has any market analysis or segmentation been performed), this process is perfectly suitable to our team. The “Agile” approach is also a necessity given the small size of our team and its limited financial resources. At the same time, the approach allows us to utilize the dynamic and fast-moving nature of our team, allowing us to focus on the code and make sure the project is feasible in the first place— rather than discover the poor feasibility of the project after a significant investment in time.

Furthermore, since our team agreed on a set of good practices for any team, we were able to skip over any inane bureaucracy and rely on principles of openness to ensure a meritocracy of ideas which does not create any barriers to enchantments or potential warnings that any member may pose. However, there are no excuses for lack of professionalism with this approach. We enlisted the communication channel, Slack, and set reasonable deadlines (set during our frequent meetings) which we must adhere to. We ensured ourselves to follow the traditional design and requirements gathering stages, so that we can create a deliverable that is consistent with our project’s vision, and so that any flaws or vulnerabilities can be exposed before any code is written.

By creating open channels for communication and work, we quickly discovered each individual’s strengths, creating a more collaborative atmosphere. Additionally, this approach allows us to avoid the problems associated with the use of “Agile” approach and avoid contractual responsibility breakdown. Our frequent “ceremonies” decide on the deliverables, which do not change, due to our commitment with this project. However, the team remains an agile entity, so that we can adapt to unforeseen circumstances. Currently, we have decided on the following sprints: research, Google Apps Scripts exploration, database development, interface development and more as necessary. We will create a burn chart to monitor each sprint.

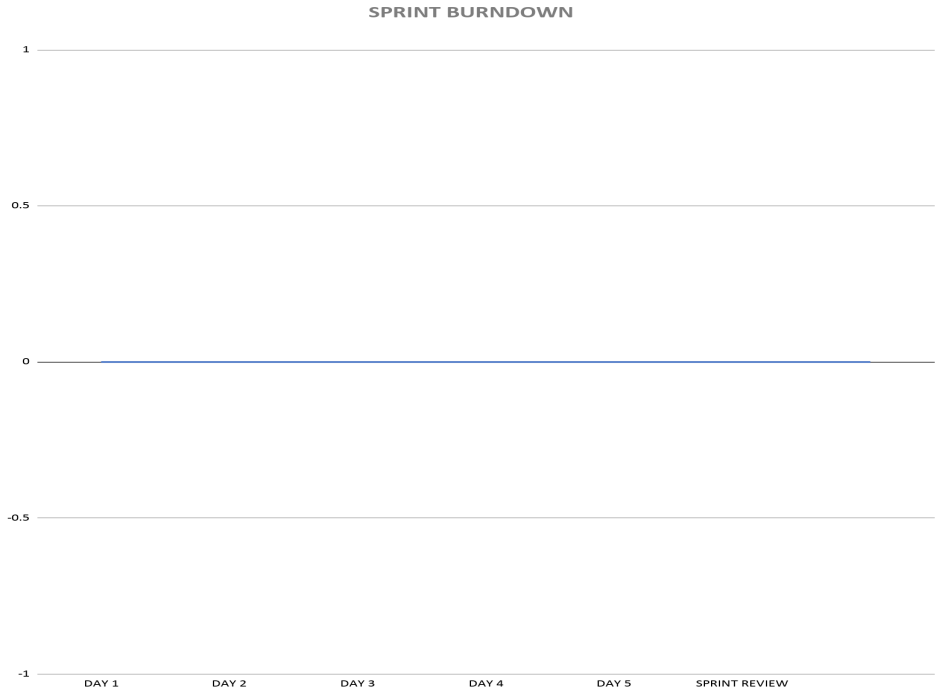
In conclusion, by utilizing a light-weight approach that suits our team’s environment and is inline with our strong commitment to this project’s final product, we hope to be successful with the “Agile Model.”

Burn Chart (Replace with Google Sheets, Graph of Sheet Data Below)

Sprint	Estimated Time Remaining			
Research	11/09	15/09	16/09	

	4 hours	5 hours	Completed	
Apps Scripts				
Database Development				
Scrollable				
Security				
Interface Development				

Note: Story points are weights assigned. “Sprint Review” is the review performed at the end of the sprint’s completion (when days left is zero), and compare the estimated time taken against actual time taken.



Requirements Engineering (SUB 3)

Q-1

User Requirements

- Custom templating feature
- Easy to click buttons
- History of templates available
- No user storage necessary
- Autofill draft message

System Requirements

- Storage on Google Account
- Database linking through Google Spreadsheets
- Scrollable or searchable templates
- Implement inverted index search mechanism

Difference between user requirements and the system requirements?

User requirements only specify the visual or interface necessities. They only mention what needs to be implemented abstractly, not how it should be implemented. The system requirements look at the details, such as what data structures and algorithms will be used to implement the user requirements. They provide a framework for the developer.

Q-2/A-2

- **Write down Functional requirements for your system.**

☐ Appearance:

The add-on shall be displayed in a pop-up window with following visual components in it:

Two Action Items:

(1) Insert Template to Email Reply

This option would allow the user to select a standard template reply from the list based on the matches found for the subject line of the reply email that the user is replying to.

This option would also allow the user to delete an existing template if it is no longer required.

(2) Create New Template

This option would allow the user to create a standard template for future use. By using this option, user will be able to populate the Template Database

The sample Add-on context menu may be as follows:

Select Action
Insert Template to Email Reply
Create New Template

(1) Insert Template to Email Reply

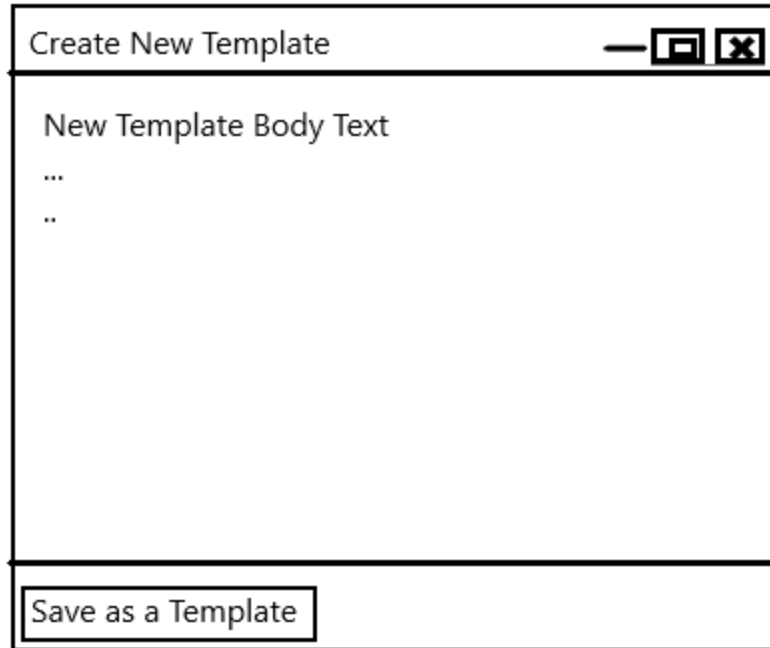
Allows user with one of two options:

- (a) Select a matched template to insert into the email being replied to
- (b) Select a matched template and delete it from the Template Database

Select a Template to Insert into Email		  
Template 1 First Line		
Template 2 First Line		
...		
		
Insert Template into Email Reply		Delete Template
Full Text of Selected Template		
...		
..		

(2) Create New Template

With this option, a user would be able to add new template to the Template Database.



Create New Template

New Template Body Text

...

..

Save as a Template

☐ Behaviour:

When a user drafts a new email, the Add-on shall allow user to save the drafted email as a template into the Template Database.

When a user replies to an incoming email, the Add-on shall allow user to select a Template from the list of templates that matched based on the subject of the email being replied.

- ☐ Upon receiving the initial focus to the “Select a Template to Insert into Email” window, the “Insert Template to Email Reply” button should only be enabled if there is at least one template available in the database (Google Spreadsheet) of the add-on. If there is no template available in the database, then there should also be an information message in the template selection window to indicate this to the user and the “Insert Template to Email Reply” button should remain disabled and grayed out. If the button cannot be disabled, then the appropriate message popup or status message should be displayed if the user clicks “Insert Template to Email Reply” button when there is no template created yet or available.
- ☐ The Template Database should have keyword fields searchable by the add-on.
- ☐ Upon receiving the initial focus to the Template Selection window, the add-on should automatically search the template database for the keywords found in the subject line of the triggering email. If a matching template is found, it should be added to the opened

Template Selection window. If there was no matching template found, then the add-on should search for keywords in the first paragraph of the email body to find any matching template. If possible, when multiple templates are matched during the search, a count should be displayed in the Template Search window's status bar with a message such as "### possible match(es)".

- ❑ The windows button marked "X" should allow the user to close the "Select a Template to insert into Email" window.
- ❑ Within the template window, all templates should be scrollable for selection
- ❑ Within the template window, any template selection should be possible by highlighting the template by positioning the focus on a particular row/record with up/down arrow key on keyboard or by left clicking the mouse/pointing device on a particular row/record. Multiple record selection is desirable but at least one selection should be possible at any time.
- ❑ Within the template window, user should also be able to select an existing matched template and use the "Delete Template" button to delete that template permanently from the Template Database.
- ❑ Upon click of the "Insert Template into Email Reply" button, the "Email Reply" window should get populated with the selected template message. If there is more than one template selected, all the template messages should be populated in the email reply body. After the template insertion, the add-on window should still remain open until the user clicks the "X" button, at which point the "Select a Template to Insert into Email" window should close with focus returned to the "Email Reply" window.
- ❑ Upon user selection of "Create New Template" Add-on option, the Add-on should open the "Create New Template" window.
- ❑ In the "Create New Template" window, user shall be able to enter a Standard Template email message text and be able to save to Template Database.

- **Write down Non-functional requirements for your system**

Ease of Use: Ideally no learning curve should be required other than a quick read of the description (available on the install page) and a brief familiarization.

Performance: light-weight in storage space needed and fast execution as consumers would expect from production ready applications

Template Size and Scalability: if the user decides to create a very large selection of the template database, then the add-on can or should limit the template match to a maximum of ten messages, then enable the function to select additional templates from subsequent matches.

Add-on reusability on operability on platform other than standard desktop: The add-on is expected to work primarily on desktop version of Gmail when used on a desktop screen, but should be usable if the user decides to access this on a mobile device in a desktop mode. If it is not possible, then the appropriate informational message should be displayed to the mobile user.

- **What is the difference between Functional requirements and Non-functional requirements?**

The difference between functional requirements and non-functional requirements is: the functional requirements states the services that the software must provide upon the particular inputs and how the system should behave, where as the non-functional requirements describes the constraints on the services or the functions that the system would present depending upon the platform available, or time constraint, security, maintainability, reliability, scalability, and performance.

Q-3

Write a structured specification of a requirement for your project. (For help: consult example on slides for a structured specification of a requirement for an insulin pump)

A-3

Gmail Add-on 3.1

Function: Open "Select a Template to Insert into an Email" window

Description: Open the "Select a Template to Insert into an Email" window when the user selects a single email from the inbox to reply to it and chooses this option from the Add-on menu.

Inputs: None

Source: User clicked on an email in his/her Gmail inbox to reply to the sender.

Outputs: "Select a Template to Insert into an Email" window displayed to user's desktop screen

Destination: Template Database Open function.

Gmail Add-on 3.2

Function: Open Template Database

Description: Open Template Database (Google Spreadsheet) that is expected to be available in the Add-on location.

If the database is not available, attempt to create a new empty database, then invoke the internal build templates function to auto-populate the template records, then resume this function.

At this point, the Template Database is expected to be available.

Continue to parse the words in the subject line of the selected email.

Inputs: Template Database (Google Spreadsheet)

Source: Opening of the Template Selection window

Outputs: Template database availability confirmation flag

Destination: Parse for search keywords function

Gmail Add-on 3.3

Function: Parse for search keywords in selected email

Description: Parse the words from the subject line of selected email.

if the subject line is empty, then parse the words from the first paragraph of the email body.

Build unique list of keywords to search in the Template Database's keyword field.

If the unique list of parsed keywords is empty, then (1) Display the information message, (2) Do not attempt to search the Template Database, (3) Disable "Select Template" button, so that user is only able to exit out of the Template Selection window.

Inputs: Selected email (subject line and first paragraph of email body text),

Template Database (Google Spreadsheet)

Source: Open Template Database

Outputs: Unique List of search keywords

Destination: Search Template Database for keyword match

Gmail Add-on 3.4

Function: Search Template Database for keyword match

Description: Search Template Database in two ways.

(1) Concatenate multiple parsed keywords found from email to create a composite keyword to be searched in the Template Database's keyword field

Find templates with composite keyword

(2) Use the individual parsed keywords found from the email to search directly in the Template Database's keyword field.

Find templates with individual parsed keywords.

Populate all template messages found into the Middle section of the Template Selection window labeled "Matched Templates Available for Selection"

Also build and maintain the array of

(a) composite keyword or individual parsed keyword along with corresponding

(b) template email text in memory or the sequence number of the records populated in the top section of "Select a Template to Insert into an Email" window. This field (b) will be used to either populate the user's reply email or to do the lookup in Template Database to get the template email text for the purpose of populating the user's reply email.

Get the count of matched templates.

If the count is > 0 , then enable the "Insert Template into Email Reply" button in the "Select a Template to Insert into an Email" window for user to select one or more template.

Inputs: Unique List of search keywords,

Template Database (Google Spreadsheet)

Source: Parse for search keywords in selected email

Outputs: Populated Template Selection window for top section.

Array of keyword to template cross-reference list in memory.

Destination: Select Template from matched templates

Gmail Add-on 3.5

Function: Select Template from matched templates

Description: When user clicks a particular template and clicks the “Insert Template into Email Reply”, button, retrieve the template email text either from memory array or from the Template Database that corresponds to the selected template.

Remain in the Select Template action to allow user to select more than one template email message, until user clicks “X” (Close Window) button.

When user clicks “X” button, close the “Select a Template to Insert into an Email” window and end the Add-on function, return the user to the reply email window.

Inputs: User selection of a template from the top section of “Select a Template to Insert into an Email” window.

Template Database (Google Spreadsheet)

Array of keyword to template cross-reference list in memory.

Source: Search Template Database for keyword match

Outputs: User’s reply email populated with the template email text selected from the matched templates.

Destination: User’s reply email.

Gmail Add-on 3.6

Function: Exit Template Selection window

Description: When user clicks the “X” button on the top right hand corner of the “Select a Template to Insert into an Email” window, close the “Select a Template to Insert into an Email” window and return to the User’s reply email window.

Inputs: User action to close the “Select a Template to Insert into an Email” window.

Source: User selection of an email in his/her inbox.

Outputs: Closure of “Select a Template to Insert into an Email” window.

Destination: User’s reply email.

Q-4

How can you use some techniques for requirements elicitation including interviews, scenarios and use-cases.

A-4

Contacting and understanding the end user needs is critical for the development of this application; as this app is created for a wide base of consumers– rather than a small set of clients. Since we do not have any clients for this project yet, we will focus on scenarios that we have encountered and can imagine. We will, however, use open interviews to open discussion and brainstorm potential implementation mock-ups. The open interviews would be scenario based, taking into account past experiences of close friends and family concrete, in order to have a set of concrete use-cases of this application.

Q-5

Check your requirements for validity, consistency, completeness, realism and verifiability (explain the concepts with an example requirement).

A-5

We conducted internal group discussion and an internal group meeting to confirm the understanding of the user requirements. So far,

- (1) The main requirement of selecting the template email message will be satisfied.
- (2) There is no conflicting requirement
- (3) Most of the requirements confirmed except the following:
 - (a) We have not finalised what should be the appropriate function when the Template Selection window is asked to be “Minimised” or “Resized” for the window operation. This item is not complete yet.

For now, the “Window Minimised” action will be ignored and will be left to the user to restore if he/she decided to minimise it.

For now, the “Window Resized” action will be ignored and will be left to the user to maintain a usable area of the window.

- (b) We have not finalised as to how a composite keyword should be formed in addition to just concatenation of the unique list of words, because this could generate a large combinations. We need to decide on how to limit the number of combination either by number of words from the unique list of words or by the number of alternate order of each word in the unique list. This item is not complete yet.

For now, the composite keyword may be searched for all words of the subject line or the first paragraph.

(4) The current requirements can be implemented using the Google’s App Scripts IDE

(5) The current requirements are verified.

Q-6

Write a sample scenario from your project and create a use case for that. (For help: consult example on slides about Scenario for collecting medical history in MHC-PMS)

A-6

What can go wrong: The Template Database file of Add-on’ disappears or did not get deployed with the Add-on. The add-on code should detect the non-existence of the Template Database and recreate one.

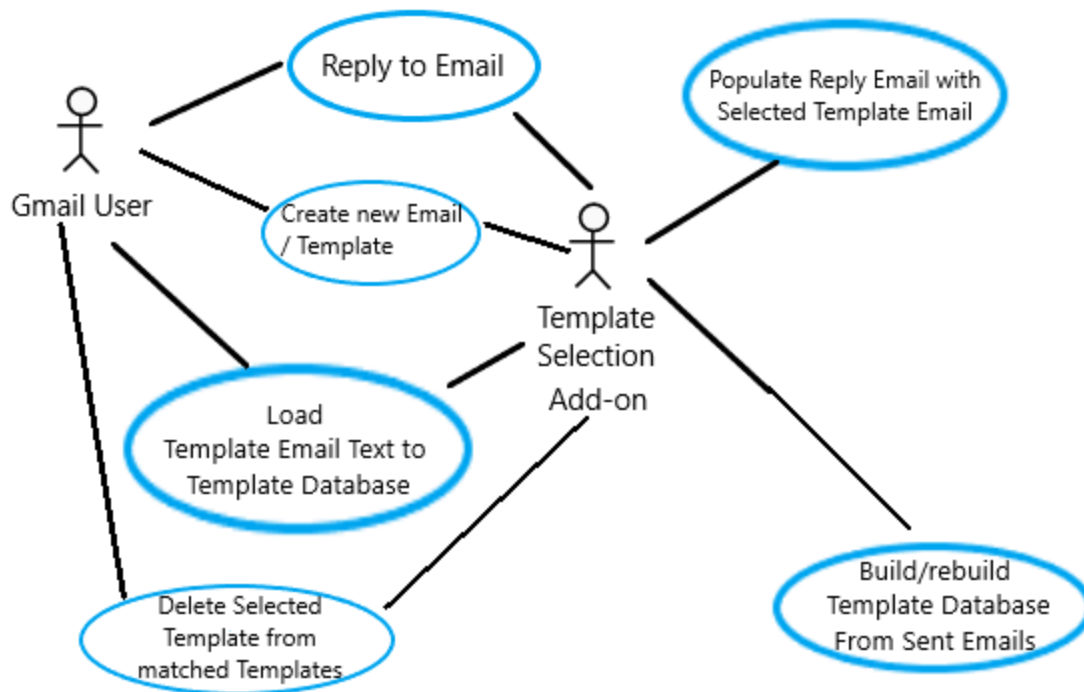
The Template Database is not pre-populated with any template email yet. In this scenario, user will see an empty Template Selection window, but will be able to exit with “Exit Template Selection” button.

Other activities: Template Database population requires a meeting with prospective user for collecting a set template email text.

System state on completion: The user’s reply email is populated with the selected template email text from one of more user selections, the Template Selection window closed and the control is returned to the user’s reply email window.

Use Case

Use cases for Gmail Add-on for Template based Email Replies



Q-7

What is Requirements management? What is the importance of Requirements management?

A-7

Requirements management is the management of continuously evolving requirements during both the requirements gathering stage and the overall software development process (unless the “Waterfall” model is chosen). As interviews continue and user requirements are translated into system requirements, modifications to the requirements are bound to occur. Thus, it is vital to create linkages between each and every requirement. With this approach, should any one requirement change, the linkages will allow one to forecast the impact of that changed requirement on all other requirements. These linkages and changes should be done in a formal, systematic fashion, as changes in requirements will also cause changes in implementation, driving both time and cost for the project completion upward.

Design Spec Misc Notes

1st iteration (no machine learning necessary)

Step 1: Template Creation by user

-->We create light Google spreadsheet with few prepopulated templates for various scenarios

-->We create a button for users to create their profiles

User sees template interface

Template interface consists of: subject form input & message form input (no authentication required since all authentication is performed by Google account services) -->Need two form input boxes so users can type their messages

User has a button to save template

-->These subject-message 'key-value' pairs must be stored dynamically into user's google spreadsheet once they hit save

Step 2: Template use

-->Create button to see templates

User has view (in the form of interface or the live google spreadsheet) and can choose the template they desire

-->Choosing a template is done ideally by having the entire google sheet mapped to the interface so they can select the template, but this is NOT necessary. We can circumvent this by having a form input so user's options must be selected by exact subject match (google Query language), and then once we get that working we can proceed to having user can search subject line with closely matching results appear in the Template Selection window.

Software Architecture (SUB 4)

Q-1 What are the advantages of explicit architecture?

With Explicit Architecture,

Stakeholder communication becomes easier as it can be used as a focus of discussion by the system stakeholders.

It helps to analyse whether the system will be able to meet its non-functional requirements or not.

In its Large-scale reuse, it allows us to reuse it across a range of systems and it allows us to develop product-line architectures from it.

Q-2 Write down the steps to choose the right software architecture?

Depending upon the type of system being developed, the architectural design process would differ. However, a number of common decisions would span across all design processes. These decisions also affect the non-functional characteristics of the system being developed.

The steps to choose the right software architecture would involve the following:

1. Identify the subsystems that make up the system
2. Identify the framework for subsystem control and communication
3. Identify major system components and their communications
4. Create abstract form of Architecture in terms of (a) Architecture in the small, (b) Architecture in the large
5. Depending upon the system being developed, get the answer to the following questions, then choose the suitable architecture for the software:
 - a. Is there a generic application architecture that can be used?
 - b. How will the system be distributed?
 - c. What architectural styles are appropriate?
 - d. What approach will be used to structure the system?
 - e. How will the system be decomposed into modules?
 - f. What control strategy should be used?
 - g. How will the architectural design be evaluated?
 - h. How should the architecture be documented?
6. Ensure that the domain concepts of all the systems in the same domain are captured and addressed in the core architecture.
7. Ensure that following system characteristics are addressed in the architecture:
 - a. Performance
 - b. Security
 - c. Safety
 - d. Availability
 - e. Maintainability
8. Determine which views or perspectives are useful in the design and documentation of the architecture.
9. Apply the Architectural patterns to determine the reuse of knowledge about the system architecture.

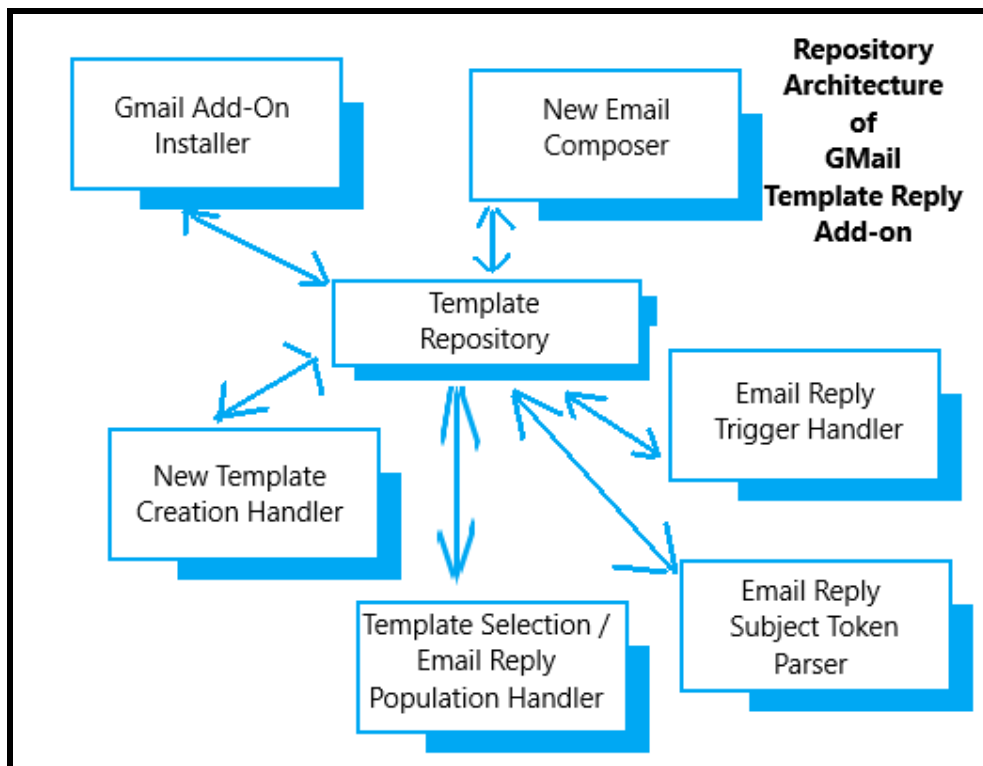
Q-3 What do we mean by the Architectural patterns? Describe briefly the Architectures and Architectural patterns covered in the lecture (all of them)? Pick the one that suits your needs (to develop your software system).

Architecture patterns are stylized descriptions of good design practices, which have been tried and tested in different environments. They should include information about when they are and when are not useful. They may be represented using tabular and graphical descriptions.

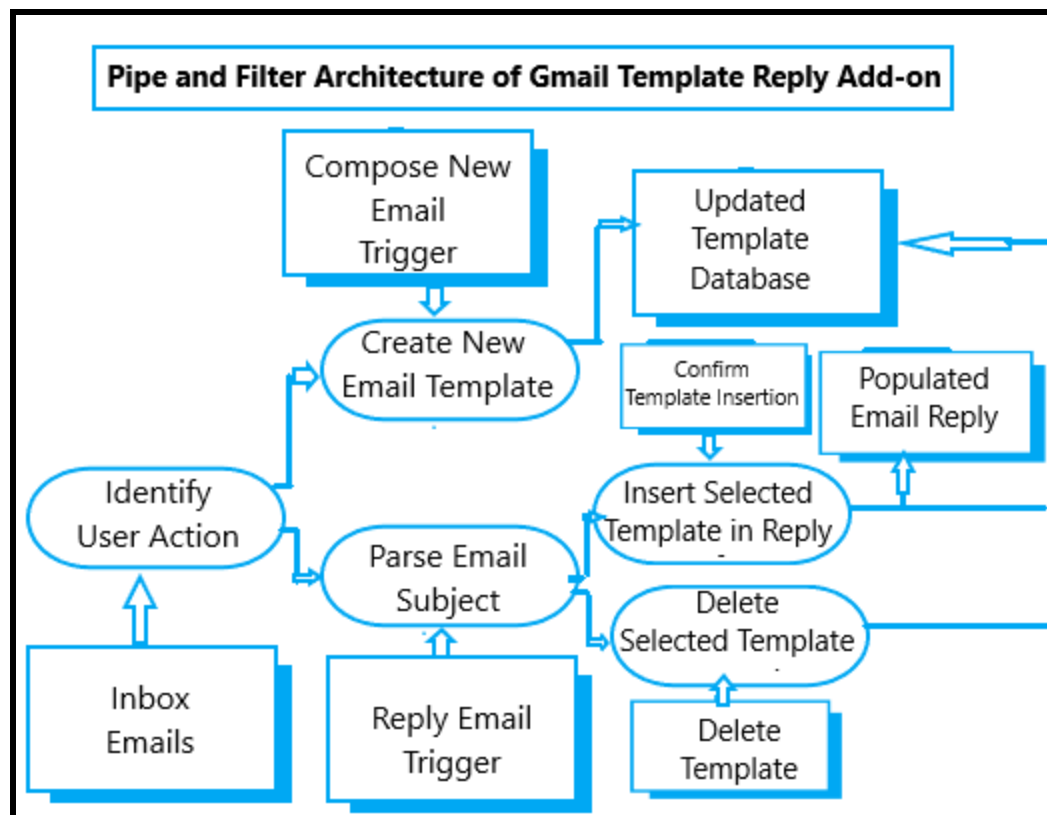
1. The Model-View-Controller (MVC) pattern: Separates presentation and interaction from the system data. the system is structured into three logical components that interact with each other. The model component manages the system data and associated operations on that data. The View Component defines and manages how the data is presented to the user. The Controller component manages user interaction such as key presses and mouse clicks and passes these interactions to the view and the model.
2. Layered pattern: Organises the system into layers with related functionality associated with each layer. A layer provides services to the layer above it so the lowest-level layers represent core services that are likely to be used throughout the system.
3. Repository pattern: All data in a system is managed in a central repository that is accessible to all system components. Components do not interact directly, only through the repository.
4. Client-Server pattern: The functionality of the system is organised into services, with each service delivered from a separate server. Clients are users of these services and access servers to make use of them.
5. Pipe and Filter pattern: The processing of the data in a system is organised so that each processing component (referred to as filter) is discrete and carries out one type of data transformation. The data flows (as in a pipe) from one component to another for processing.

For the purpose of the development of our system, we will be using the MVC pattern since our system will be providing the functionality to an application that is web based application, so MVC would be most suitable for it.

Q-4 Draw a repository architecture of the proposed system (your project).



Q-5 Draw a pipe and filter architecture of the proposed system (your project).



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UPDATE on Application Window Mockup

As the construction activities phase progressed, we realized two application windows were necessary. The UI of the Gmail Template Engine now consists of two main application windows, one for creating new templates (launching the compose UI of Gmail), and one for using the add-on while in a “reply” state (where the user has already selected a recipient to whom they are replying).

Both Application windows were created using the mockup below as a design inspiration:

Create a Reply Template

Create Delete

☐ Radio Button

☐ Radio Button

☐ Radio Button

☐ Radio Button

☐ Radio Button

Sub 5

Q-1 What do we mean by Middleware technologies? What type of services does middleware provide?

A-1

Middleware is a pipeline or channel that acts as the medium on which clients and servers can be linked. Middleware technologies include a wide array of software and services which connects front-end programming (user-interfaces) with back-end programming (server-side, hidden from the user) in the distributed computing. Middleware provides the following services for connecting software:

1. Business Process Architectures
2. Message Brokers
3. Application Servers
4. Transport

Q-2 Provide names and a brief a description of all Middleware technologies covered in the lecture. Compare them to conclude this question.

A-2

- Common Object Request Broker Architecture (CORBA) is a group of synchronous middleware technology which provides naming, notification and transactional services.
- Message Oriented Middleware (MOM) is a group of asynchronous middleware technologies which can provide asynchronous communication between processes, application and systems. Furthermore, it can facilitate “Send-And-Forget” requests, where only delivery without confirmation is acceptable. MOM technologies can also provide atomic transaction processing and persistence mechanisms to survive server failure.
- J2EE provides execution environment for J2EE specific components such as “EJB”, “Message driven beans” and “Connectors.” Through EJB containers, J2EE provides, transactions, security, directory, threading and connection pooling for efficient use of server resources.
- Message Brokers can provide fast message processing and high level programming features for exchange, manipulation and routing messages between an app’s components.
- Business Process Orchestrators provide support for work-flow type of applications by describing, executing and managing the intermediate states of business processes.

Q-3 Are you using any middleware technology for your software system? If yes, how have you picked this? If no, why you don't need any?

A-3

The programming environment we are using is Google's native Apps Scripts, which has its own dedicated programming language (based on JS) and its own dedicated IDE. With all communication done directly through Google's services, all middleware is implemented behind-the-scenes by Google. For one portion of our project, we do plan on using an external API, provided by Google, which will link scripts written in the Python programming language. Depending on the implementation we use, middleware such as Google's AuthMiddleware() technology may be required. In that case, our documentation will be updated with the details of this middleware technology. For our project, we have not voluntarily chosen any middleware, but rather rely on our platform's documentation to instruct which specific middleware is required. As a result, in one sense, Google's development platform itself can be viewed as middleware. Google Apps Script acts as a middle ground, which connects Google's server-side programming (encapsulated behind Google's own programming language) with Google's front-end services (such as Gmail, Sheets etc.). Many functions have high-level functionality, and abstract away the lower-level details so that programmers can focus on high-level implementation. Developers using the Google platform like ourselves, never develop directly on the front or back end, but through Google's provided interface and language.

Sub 7

Q-1 How will you decide about Building or buying a software module/component?

In today's age, off-the-shelf software solutions are ubiquitous for a variety of services. Websites/ web apps of businesses ranging from sole proprietors/freelancers to multinational corporations all use common components. For example, freelancers asking for donations and retailers both implement PayPal, Google Maps API, AWS etc. Many software products (and their competitors) have become industry standards which can be customized to your liking. AWS offers tailored resources, databases for fields of medicine or law are cheaper to outsource rather than develop in-house and many tools are free to use completely (such as Google's Apps Scripts' tools which you can tailor using programming as we are for our term project). However, there are some cases when its necessary to build your own components. For instance, consider our term project, templating features are available by Gmail itself or through add-ons, but they are hidden away and not convenient for users. In this case, it makes sense for us to develop our own solution.

Similarly, for organizations in the intelligence sector or other high-security sectors, it is not sensible to outsource or purchase tools that may have data intercepted through cloud resources or even to the company providing the component itself. In such cases, tailored applications are not, the data flow simply cannot leave the organization. Also, industry standard software modules are not necessarily always cheaper. Wix , GoDaddy, etc are all sites that allow you to build your own website, but these economical options usually lack robust server-side programming and its often cheaper to hire a

developer to this instead. Many services seem cheap at first but building them can mean costs saved over the long term. Moreover, even tailored processes require a learning curve and the task of tailoring requires investment. In some cases, a team of developers can build a component with the same time/money required for tailoring one (when time of cross-shopping, researching code etc. are compared).

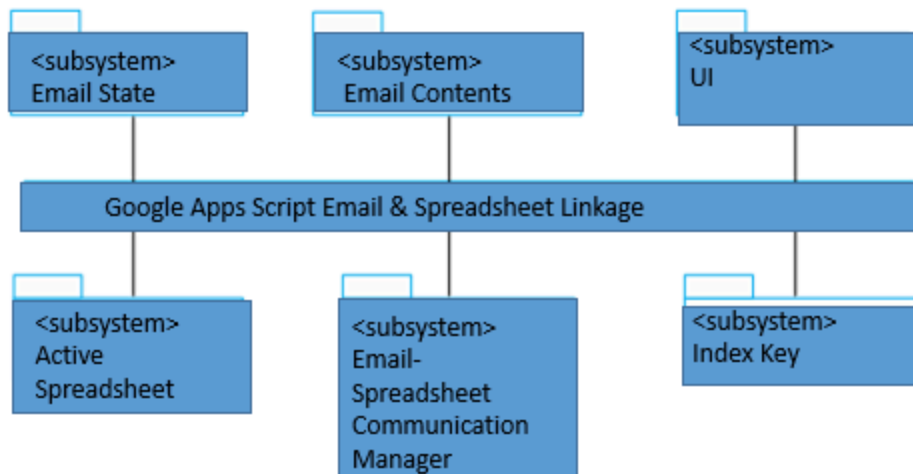
Q-2 What is the impact of system context and interactions on the software architecture.

System context tells us what features the solution will be responsible for and what it won't. Rather, more accurately, it represents the scope and domain where the solution will reside (not very different from the English definition of context). System interactions are the relationships between the solution and its environment. By assessing the system interactions and context, we come to understand the boundaries of the system. Both of these concepts can also be modelled as more practical definitions. A system context model is a structural model which shows the other systems in the solution's environment.

A system interactions model is a dynamic model (Recall that software requirements are constantly evolving due to changing user requirements and updated system requirements once the construction phase of the process model begins. As a result, interactions with the system are also always changing and thus are represented as a dynamic model of the interactions of the solution as it will be used by end users.

Q-3 Identify the major components that make up your software system and their interactions, and then organize the components using an architectural pattern such as a layered or client-server model.

Major Components of Gmail Template Engine



Q-4 How can you identify object classes for a software system?

We can identify object classes using four strategies. The first strategy we can use is to carefully analyze and interpret the grammar used in the requirements document. Secondly, we can try to identify the physical objects which will become software objects. Thirdly, we can interpret the behaviour that the different objects should have. Finally, we can create mock scenarios in which we analyze the objects, attributes (states, characteristics etc.) and methods (based on behaviour) that arise in each scenario.

Q-5 Identify object classes for your software system.

(Object classes below, see methods explained in class diagram below for further clarification)

Email_State //read and change email state from default to compose

Read_Email_Contents //parses and stores subject line, recipients and email body separately

Spreadsheet_state //Create one spreadsheet (for add-on use) for users (if not pre-existing). Spreadsheet also has a state, add-on spreadsheet should become active when add-on is in use.

Extract_spreadsheet_info (similar to “Read Email Contents”)

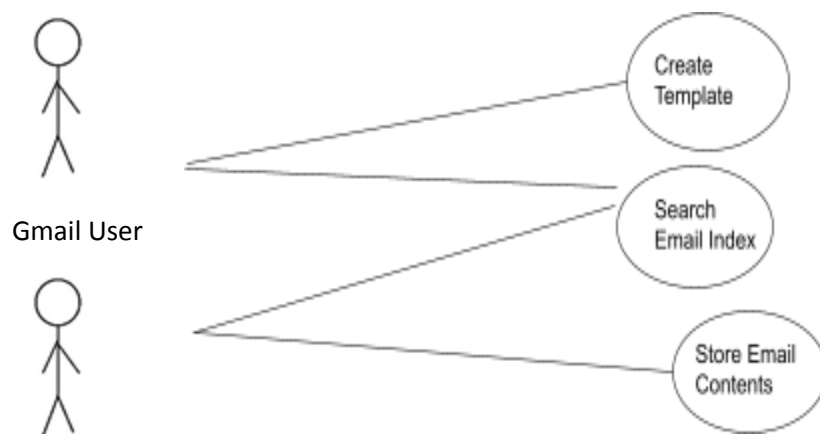
Q-6 What is a use case diagram?

A use-case diagram is a layout of the actors, business system and their abstract interactions. It is, as the name suggests, a diagram of the different classes of usage which are inferred from the requirements engineering stage of SWE. Each actor is represented as a person (drawn as a stick figure), and every actor is involved in particular scenarios. Actors range from end users (i.e. customers, employees, managers) to even non-humanoids such as database tools (i.e. external systems) or physical environments (ex. weather). However, the software solution itself cannot be an actor, since use-case diagrams model scenarios of use of the software solution. Each interaction is categorized and drawn as a bubble to which actors are owners of (this information is later used in swim-lane diagrams). Use case textual descriptions take use-case diagrams one step further by adding six key pieces of information:

1. Unique name
2. Participating actors
3. Entry conditions
4. Exit conditions
5. Flow of events
6. Special requirements.

Q-7 Draw use case diagrams for your system?

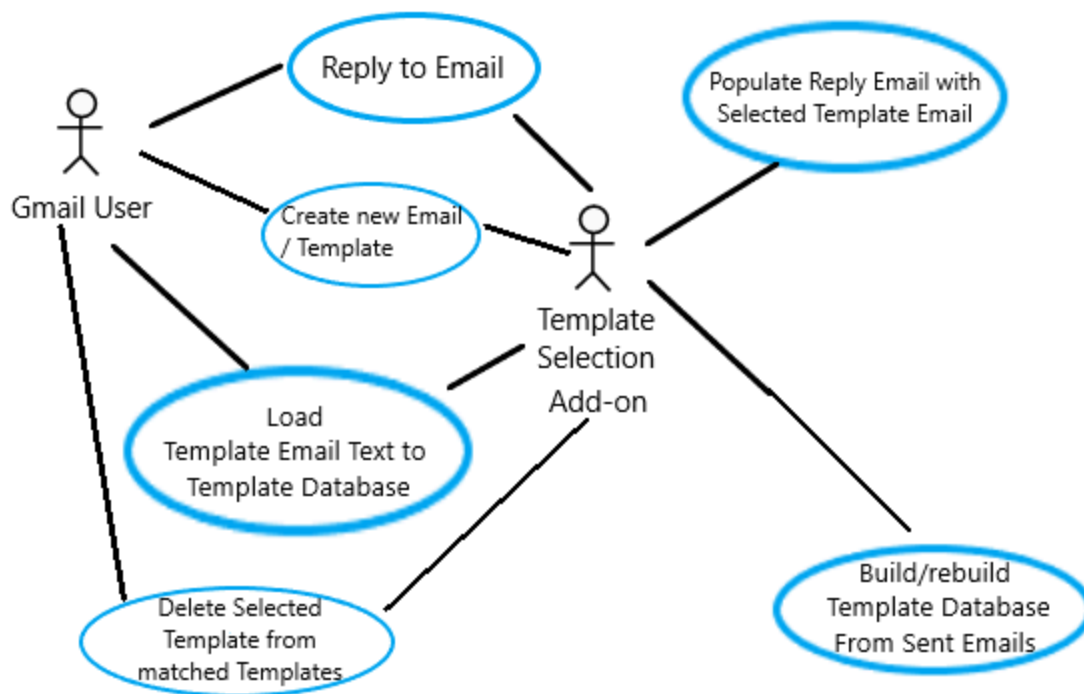
High-level abstract use-case diagram:



Spreadsheet Tool (external system)

High-level precise use-case diagram:

Use cases for Gmail Add-on for Template based Email Replies

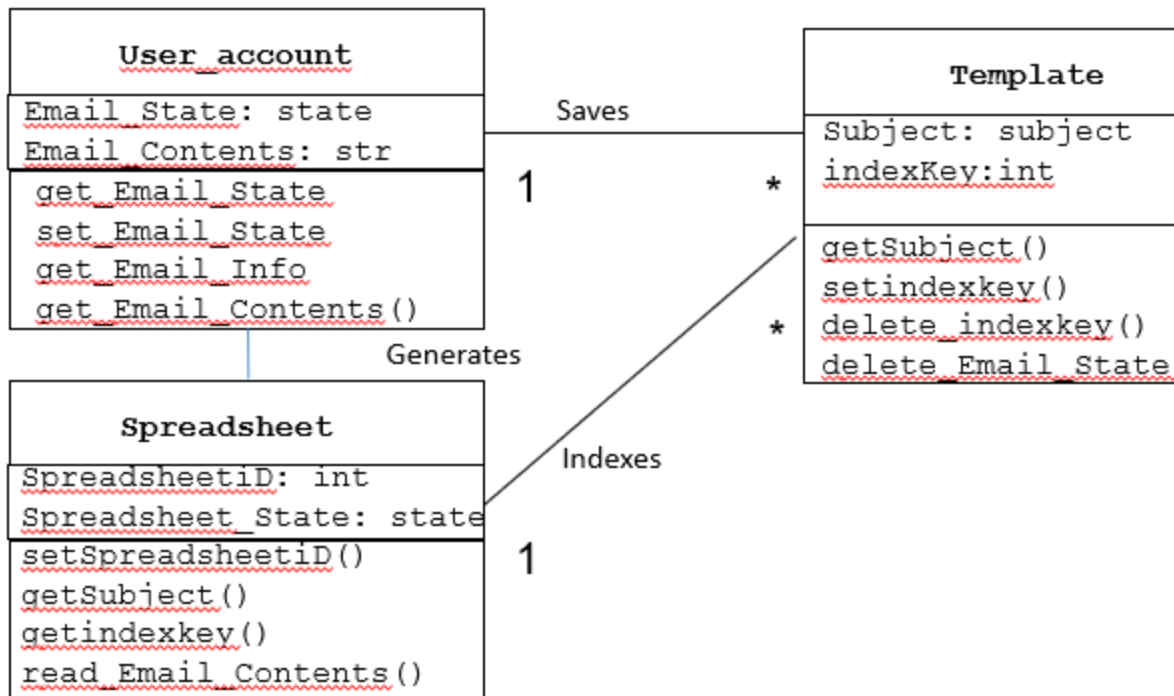


Q-8 What is a class diagram?

Class diagrams are much less abstract than use-case diagrams or textual use case descriptions. They no longer model flow of information or scenarios which model the user interactions with the software solution. Rather, they model the implementation of the software solution. Abstract actors are replaced with classes which model the actors but are also entities in the software solution. Unlike use-case, activity or swim-lane diagrams, class diagrams are mappable to the software solution in a one-to-one relationship. Each class represents a concept (identified with a unique name) and includes data fields. Data fields consists of attributes (i.e. states of different type) and operations (i.e. method signatures). Class diagrams can then be expanded on to include instances (of the class, with underlined names to differentiate from the class the instance belongs to) with the attributes now specifying the values that instance holds.

Q-9 Draw class diagrams for your system?

Class Diagram for Gmail Template Engine



Q-10 What do we mean by Implementation issues (Reuse, Configuration management, Host-target development)? How will you handle them?

Reuse: Most software problems occur commonly and are shared between numerous solutions. Reuse existing code whenever possible as existing code does not need to adhere to the same strict testing procedures because it already has proven its reliability.

Host-target development: Seldom do we develop on our clients' machines. Rather we work in a separate environment from the one that end users operate on. Teams must be aware of any limitations and possible restraints or complications of executing on client machines.

Configuration management: During construction activities, one must strive to track and maintain multiple versions of the development process, you have to keep track of multiple versions of each software component in a configuration management system.

10 b) How will you handle them?

Reuse

We will handle reuse by continuing to reuse the tried and tested methods that Google Apps Scripts provided. Furthermore, we will import code that has been tested from our other projects, such as search and front-end UI technologies whenever possible.

Host-target development

Fortunately, host-target development is an issue which we considered early on. Each of us created separate emails for the strict purpose of testing our add-ons execution on end-user environments. Due to the nature of online Google accounts, Google encapsulates all features' functions and handles all processing of their features so no additional considerations of client machines are required. Although we will be vigilant, if the solution works for our testing accounts, they will work for all end-user Gmail accounts.

Configuration management

Again, we were cautiously aware of this issue early on. Fortunately, Google's versioning feature (which applies the concept of version management learned during the lecture), which saves each change to a project as separate checkpoints which can be reverted to at anytime of the process. As an additional contingency, our separate Gmail accounts (created for testing, as explained above) will house individual components for a modular maintenance performed by each account holder of the team.

SUB 8

Q-1 Write short notes on the following:

Synchronous message: Synchronous messages are messages which depend on a sequential process. Consider the following scenario:

An event is fired. A message is sent in association of this event to another component. The component then sends a message to the server. Each message is sent sequentially and is served based on some queue management (FIFO, priority queue etc.). As each message is sent, the sender must wait for each response and cannot send another message until a response is received. Synchronous messages are a form of two-way communication as users can send and receive messages (from the server).

Asynchronous message: Asynchronous messages implement a "Send and Forget" protocol. Rather than have external components wait for previous actions performed by preceding components, asynchronous messages are sent without respect to any ordering. Asynchronous messages are a form of one-way communication (as you do not wait for responses, rather only can send messages)

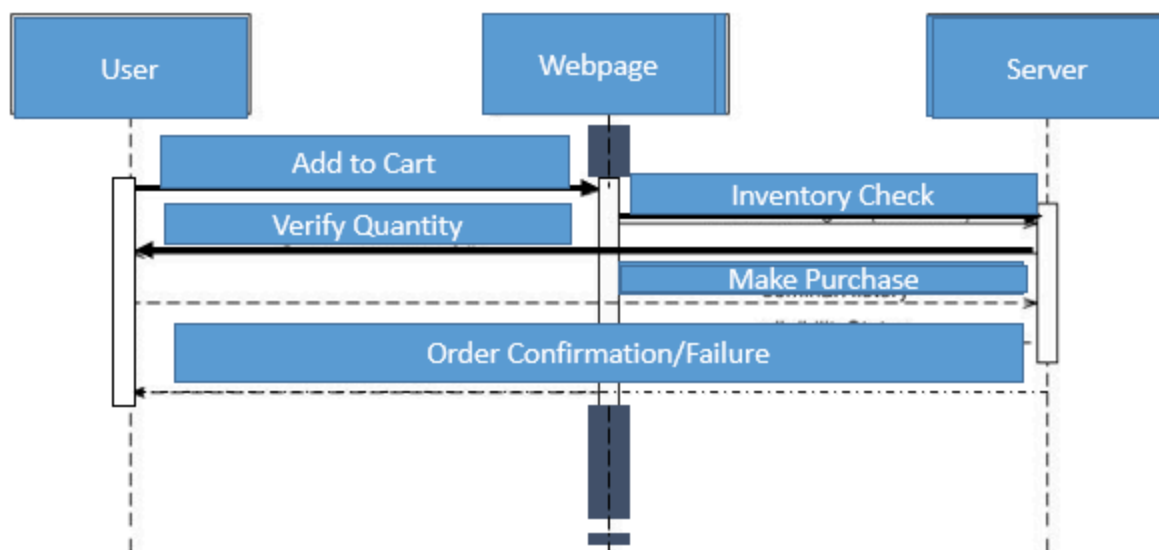
Difference between Synchronous and Asynchronous messages: The execution of synchronous and asynchronous messages differs by their method or control of response. Synchronous messages must wait for responses of previous messages before sending another message. Asynchronous messages can create backlogs of messages, accepting new messages while previous messages are responded to. An advantage of asynchronous messages is the allowance of running concurrent processes regardless of whether or not they have interdependencies. Although synchronous messages allow execution of concurrent processes, they do not allow for concurrent execution of interdependent processes. In summary, synchronous messages rely on two-way communication (sending and receiving within fixed time intervals, controlled by a clock) whereas asynchronous messages only permit sending of messages (without relying on fixed time intervals between messages).

Q-2 What is usage of Sequence diagrams?

Sequence diagrams are used to show an order of events. They can model flow of data through sequences of events, happening chronologically or as concurrent processes. Flow of actions are modelled as they pass certain events (points in time) that occur during one execution of a particular usage of the software. One iteration of a sequence can represent one activity that takes place, such as one transaction, database update (inventory control), etc.

Q-3 Draw Sequence diagrams for your system/or picking any example (should not be very small).

Sequence Diagram for Order Transaction



//Note: All messages are synchronous

Q-4 What is usage of State Machine diagrams?

State diagrams are used to provide an abstract description of a system's behaviour. It displays what events cause a change of state in the system. The same events can cause different responses depending on what state the system is in. A graphical representation of states can make a system's behaviour easier to understand.

Q-5 Draw a state machine diagram for Toy Train

Initially, headlight is off, and train is not moving.

If power on, headlight shines, and train moves forward.

If power off, headlight goes out, but train still moves forward.

If power on, headlight comes on, and train does not move.

If power off, headlight goes off, and train stays stopped.

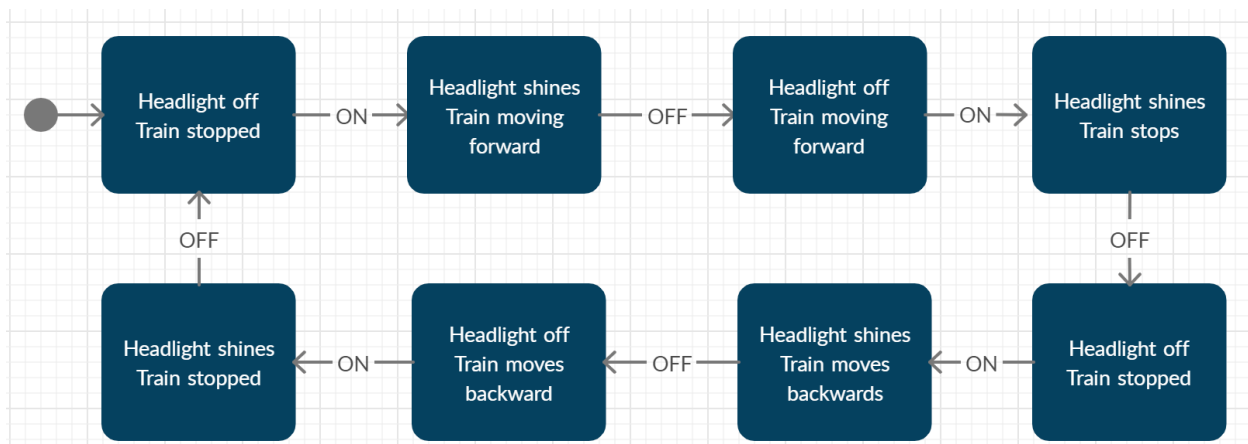
If power on, headlight comes on, and train moves backward.

If power off, headlight goes off, but train still moves backward.

If power on, headlight comes on, and train doesn't move.

If power off, headlight off, and train remains stopped.

Repeats from step 2 above.



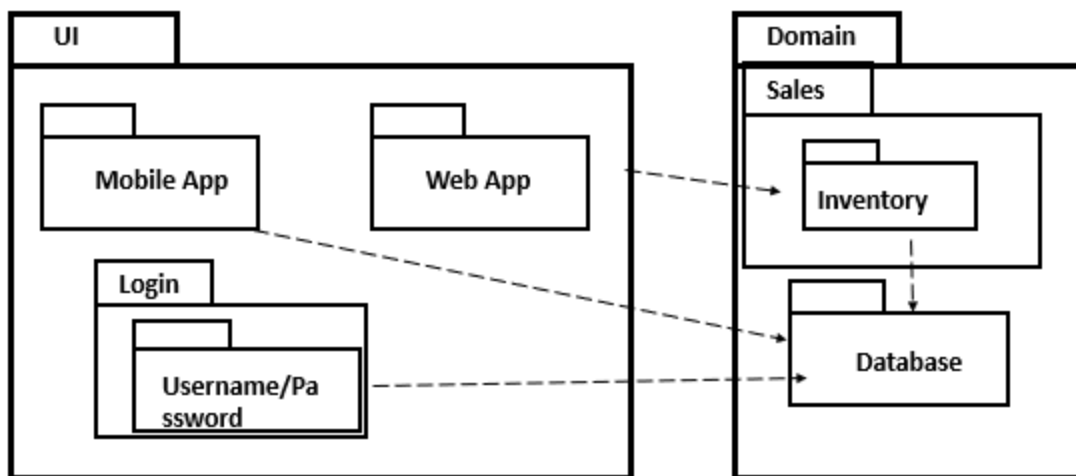
Q-6 What is usage of Package Diagrams?

Package diagrams are used for modelling UML software components and modules. It allows mapping of core functionalities as directories (similar to a file system for object oriented style models). A practical example of readily available package diagrams can be found in GitHub's file layouts.

Q-7 Draw Package Diagrams for your system/or picking any example (should not be very small).

Package Diagram: Lab 8

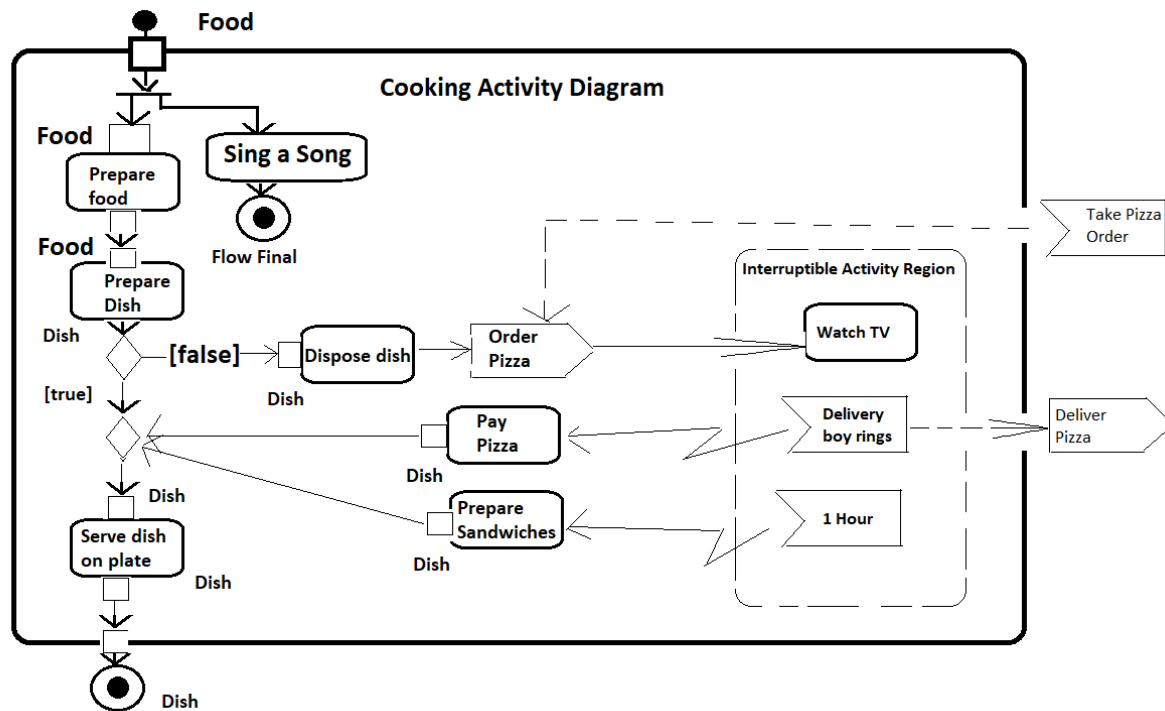
Small Business Ecommerce Solution



Q-8 What is usage of Activity diagrams?

Activity diagrams are used to map the relations between different actors of the system. They model the interactions of the different actors and the activities that they can partake in within the system. Activity diagrams can emphasize the flow of sub-activities between actors as activity nodes. Activity nodes can have different edges (control, object) to model the different types of completion flows or passing mechanisms, etc.

Q-9 Draw Activity diagrams for your system/or picking any example (should not be very small).



SUB 9

Q-1 Write a brief description for all software testing techniques covered in the lecture (where possible add examples).

Regression Testing is mainly used for testing between incremental changes. It is suitable for testing where multiple versions of the solution occur and tests must be conducted after each version update.

Release Testing is testing ideally by a test team which is independent from the development team. This is testing performed at its final stages. Inputs are provided and matched against expected outputs. Release testing is a type of system testing, not unit-testing. Although this testing also uses black-box testing, it should not be confused with unit-testing. Test-cases are not being used as unit-testing– rather testing is done from a user's point of view (touching source code is forbidden). The test team will test all possible inputs to see if any input "breaks" the solution.

Requirements-based Testing is the process of creating requirements (or using select, testable functional requirements from the software requirements document) which serve as test-cases. Those

Performance Testing

User Testing is Three types exist. In Alpha-testing, users are contacted and are asked to test the software with the development team. Testing can be in front of the team ... or remain in contact....

Acceptance Testing is ... in agile acceptance testing happens at each increment. With other software methodologies, acceptance testing only happens at the end, when customers are involved.

Unit Testing

Inspection Testing

Object-Class Testing

Validation Testing

Automated Testing

Unit Partition Testing

Interface Testing (incorporates stress testing)

System Testing

Program Testing

- Validation and Defect Testing

- Inspection Testing

Object-Class Testing

- Test all operations (methods) of objects

Validation Testing

- A successful validation test shows both developers and system customers that the solution performs all operations it was intended to execute

Unit Testing

- Unit Testing employs test case-based testing methods

Automated Testing

- Unit tests are automated without manual intervention

Partition Testing

- Input data and output results fall into different classes

- Test cases should be chosen from each partition

Interface Testing

- Detect faults due to interface errors

- Interface misuse, misunderstanding, timing errors

System Testing

- Components of a system are tested as a whole focusing on interactions between them

Use-case testing

- The use cases developed from the system are tested
- Use sequence diagrams to test

Regression Testing

- Test if new changes haven't broken previously working code
- Better if automated

Release Testing

- testing ideally by a test team which is independent from the development team. This is testing performed at its final stages. Inputs are provided and matched against expected outputs. Release testing is a type of system testing, not unit-testing. Although this testing also uses black-box testing, it should not be confused with unit-testing. Test-cases are not being used as unit-testing— rather testing is done from a user's point of view (touching source code is forbidden). The test team will test all possible inputs to see if any input "breaks" the solution.

Requirements-based Testing

- is the process of creating requirements (or using select, testable functional requirements from the software requirements document) which serve as test-cases. Those

Performance Testing

- Test performance reliability of the system
- Use profiling and metrics
- Stress testing

User Testing

- is identifying a small functionality, write code, then run test on it. Object or method.

Three types exist. In Alpha-testing, users are contacted and are asked to test the software with the development team. Testing can be in front of the team or customers

must remain in contact

2. Write 20 Requirements tests for your system. (Write a requirement followed by a test for that requirement)

Requirements 1-10 (Automated Testing & User Testing based test cases)

Requirement	Test
View Email	Automated test function to check if email is parsed
Read Subject Line/Body	Automated test function to check if the subject line and body is NOT NULL
Set Subject-Only Template	User saves subject line of template, checks if GUI if subject is visible
Store Current Email as Template	User has button to allow storage of current email. User can inspect GUI to check for newly added email.
Store Email Subject, Recipients, Body in Spreadsheet	Spreadsheet must have a one-to-one mapping of each email in template
Enter valid input for product code	Product code is checked against product database and warning is issued if product code is not found
Enter valid input for work order	Only work orders present in database matching date of entry of particular threshold (issued by supervisors) are accepted. Otherwise, system declines input and a warning is issued.

Calculator requires valid input for arithmetic operations	Input for calculator: User performs divide by 0 operation. Warning is issued "Math Error Cannot Divide by 0"
Calculator requires valid input for trigonometric functions	Input for calculator: Warning issued if tangent function (and its variations) is fed values of pi (or variations of pi when tangent function is modified)
Calculator requires valid input for exponential functions	Input for calculator: User may not input 0 or negative integers (or decimal numbers) into logarithmic functions. Logarithmic bases must also be positive, non-zero

Requirements 11-15 (Access & Authorization based test cases)

[Ved] Requirement : If a user has access to a door, the key should validate successfully

Tests

- Setup a user without access to the door, the key should fail to validate with an error message
- Setup a user with access to the door, the key should validate successfully and the door indicator should be GREEN
- Add a user to a door, a record should be created granting the user access to that key within the database

[Ved] Requirement: Only an admin user should be able to add or remove another user

Tests

- Create a user without an admin role. Attempt to add another user, it should fail with an error message.
- Create a user with the admin role. Attempt to add another user, it should succeed and a

database record should be created.

- With an admin user, attempt to delete another user. The user and record should be deleted.
- With an ordinary user, attempt to delete another user. The user and record should not be deleted.

[Ved] Requirement: If a user leaves an organization, all keys should be removed automatically

Tests

- Add a user to an organization. Assign multiple keys to that user. Remove that user from the organization. The keys should now be removed and all associated database records should be deleted/modified.
- Add user1 to an organization A. Assign multiple keys to that user. Add user1 to organization B. Assign multiple keys to that user. Remove user1 from organizationA. organizationA keys should be removed. OrganizationB keys should not be modified.

[Ved] Requirement: If an access point is removed from the organization, all associated keys should also be revoked.

Tests

- Add an access point to an organization. Add a user to that access point and organization. Remove the access point from the organization. The associated key should be removed, and all database records should also be removed. All other keys should not be removed.

[Ved] Requirement: User should be able to login to the mobile app

Tests

- Create a user with credentials in the database. Input the credentials in mobile app. Validation should pass if credentials match. User is granted with logged in state.
- Same as above, but if credentials do not match -- user is granted with an error screen and not given any further permissions

Requirements 16-20 (Test Cases for Email Add-ons involving templates & databases based on

chronological user actions)

Requirement 16. The email add-on should be displayed as a pop-up with following two action items:

(a) Insert Template to email reply

(b) Create New Email Template

Prep steps: Open Gmail

Install the developed add-on s/w product

Go to Inbox pane of user's Gmail.

Tests : Select any email from the Inbox

Expect: The email add-on pop-up must be displayed flushed to the right edge of the window

The pop-up should have two menu items as specified above as (a) and

(b)

Both menu items must be clickable buttons

Requirement 17. Upon selection of Insert Template to Email Reply menu item, the user should get a list of available templates to choose from or get an Informational message if there is no template to choose from

Prep Steps: With the add-on enabled in user's Gmail, select an email, to cause the email add-on pop-up to open, then select "Insert Template to Email Reply" menu item

Tests: The window must be populated with any available template.

In case of no template available to choose from, an informational message must be displayed such as "There is no template to choose from"

The window must be closable.

If there are templates available to choose from, then all the listed template must be selectable.

If there is no template available to choose from, then user must be able to close the pop-up window, to return to the selected email.

Upon selection of an available template message in the top pane of the pop-up window, the full message must be displayed in the bottom pane of the pop-up window with scroll bar if the message is longer.

Upon click of the “Insert Template to Email” button from within pop-up window, the selected template must be inserted to the selected email, then the pop-up window must stay open for user to have option to select any additional templates or close the pop-up window.

Requirement 18 . User must be able to create new email templates to save in the template database

Prep Steps: With add-on enabled in user’s Gmail, click “Compose” from Top-Left corner to draft a new email. The add-on popup should be activated with two options, select “Create New Template” menu item

Tests: The pop-up window must display the screen with title “Create New Title”, an email composer/text editor window and “Save as Template” button.

The cursor should be in the email composer/text editor window for user to start entering template text.

The “Save as Template” button must be clickable.

If the user tries to save a template without any text in the email composer/text editor area, then the user must get an Informational message at the bottom of the pop-up window.

If the enters some text in the email composer/text editor window, then clicks the “Save as Template” button, the template must get saved in the Template Database, with an Informational message to the user.

After the Template is saved, close the pop-up window, then from user’s Inbox, select an email and click “Reply” to go back to template pop-up window, then select “Insert Template to Email Reply” menu item. The newly added template must be displayed in the list of available templates.

Requirement19. The user must be able to delete a previously created and existing template

Prep-steps: The delete template function is accessible from the “Insert Template to Email Reply” menu item. So, to access the delete template function, select an email from user’s Inbox, then from the opened email template add-on pop-up menu, select “Insert Template to Email Reply”

Tests: The list of available templates must be displayed in the top pane of the pop-up window. The displayed templates must be selectable from the top pane.

Upon selection of a template in the top-pane, the full text of the email template must be displayed at the bottom pane of the pop-up window.

Select a template, then click the “Delete Template” button to delete the template from the Template Database. The Template must get deleted from the Template database and the display in the pop-up window must reflect the removal of the template.

Requirement 20: The Template selection function must match words from subject line of the selected email.

Prep-steps: From the user’s Inbox, select an email to reply to, from the pop-up window, select the “Insert Template to Email Reply” menu option.

Tests: The list of displayed templates must contain one or more words found in the subject line of the email being replied to.

If there is no matching template, the user must receive an information message that there was no matching template to choose from.

Submission 10

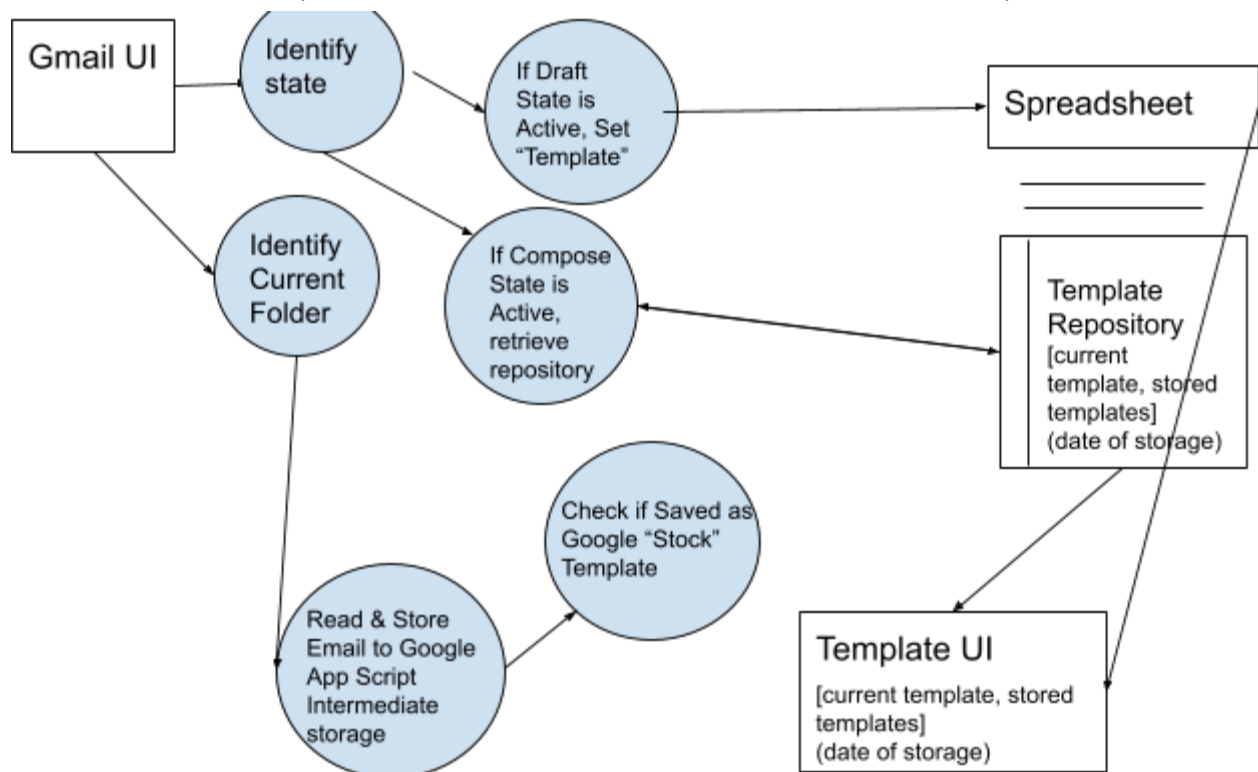
CCPS 731 -

Q-1Draw DFDs for your system.

[Hemanshu, for our template project]

DFDs are diagrams which show all data flow in and out of various processes. Unlike flowcharts, processes do not follow any sequential ordering, and do not imply any ordering whatsoever. Additionally, DFDs allow the diagramming of concurrent processes (unlike flowcharts), but they do not differentiate the time taken by each process. One can interpret the time taken for different processes, but the diagram itself does not differentiate between processes, rather it models processes of varying lengths as if they

were equivalent in duration. The objective is to only model data flows between entities, storage structures as well as the processes used as mediums of data transformation or transportation.

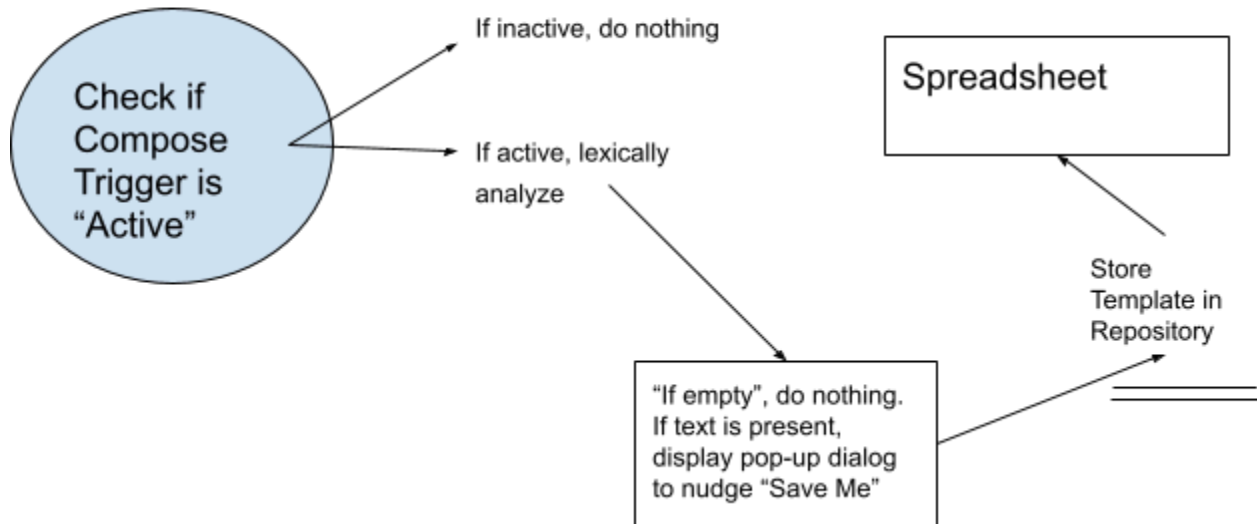


Q-2 What is a DFD Fragment? Draw DFD Fragments for your system.

[Hemanshu, for our template project]

A DFD fragment is a subsystem of the original "Data Flow Diagram", but now focussed on one singular event. Events are defined prior to the drawing of DFD fragments, and are based off of events from the event table. DFD fragments, like DFDs, are capable of showing concurrent processes (unlike flowcharts), and are limited to a single response to the chosen event, within a single process symbol. Likewise, only data stores related to that single process are shown, and any other related data stores inherited from other processes are omitted.

DFD Fragment for Composing an Email

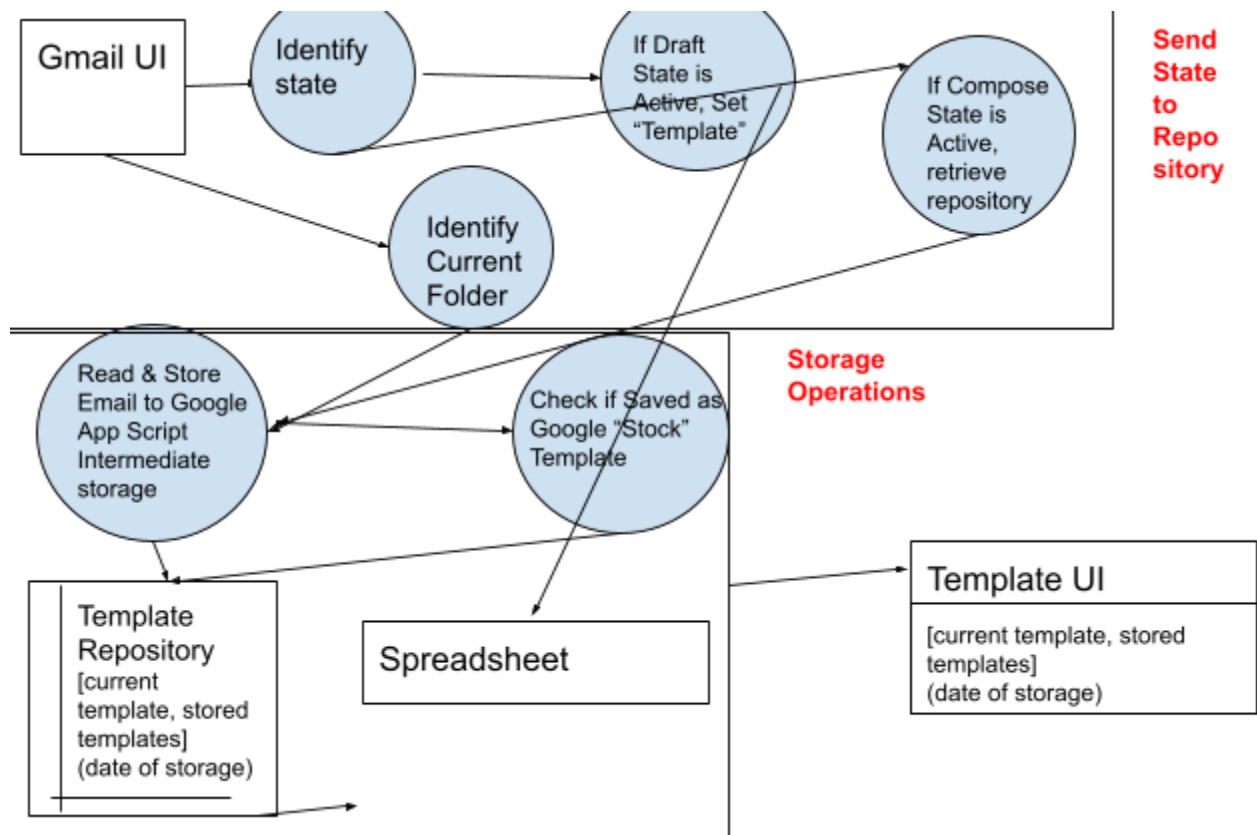


Q-3

Draw Context Diagrams for your system.

[Hemanshu, for our template project]

Context diagrams are similar to DFD fragments, as they isolate the DFD to one process at a time, however context diagrams instead aim to show the boundaries of the process in isolation. They add information not present in the DFD fragment, such as including related entities and data structures that the DFD fragment omits. Of course, context diagrams also show the highest-level view of the entire DFD, so it encompasses all processes in the system, abstracted to different levels and "split" by process. Context diagrams illustrate the context of each process, and where each flow of data begins/ends, merging the context of each process/abstractions.



Q-4

What algebraic notations are used to represent the data structure? Provide a few data structure examples for your system.

[Hemanshu, for our template project]

The algebraic notations used to represent data structures in our “Templating Engine” project are [] for either/or storage options and () for optional storage parameters. Once all of the data flows have occurred, the template repository stores all information by default, including date of creation, date sent/received, frequency of selection, subject line and body text. However, during template creation, subject-line elements and all date elements are optional– it is up to the user whether they wish to sort their templates via these parameters (they are all captured by default, but are stored at the discretion of the user). We also make use of [], to denote that a template can either be written to the repository in real-time, or can exist as a template before use of the add-on. All templates during storage are connected via the compose trigger, and all compose actions trigger the add-on so that the email can be stored as a template. However, it is impossible for a template to be written into a data structure while another template is being retrieved. Stored templates can be modified, but the “engine” cannot perform two of these operations at once, at least not for the same draft email. (We rely on Google’s middleware concurrency mechanisms to handle running two or more instances of the add-on. We assume the add-on will be able to perform varying operations for each instance of a draft email, however this is yet to be tested). We do not use the repetitive action notations because the objective of the add-on is to be

precise and sensitive to templates varying by a single character, so all data is stored in an atomic fashion. The add-on itself is created to handle repetitive actions for the user, but the architecture of the add-on does not rely on repetitive actions.

Q-5

What is a Structural Record? Provide Example of Structural Record for your system.

[Hemanshu, for our template project]

A “Structural Record” is an entity which stores groups of fields in a hierarchical or logical form. Structural records may be nested within themselves, and are often used for storage of fields such as entity attributes (such as the contact information of actors within a system, ex. employees, patients etc). Structural records must be specific, they cannot hold “placeholder” or instructional information, only records related to the business case are allowed.

Ex. for “Template Engine” Project

recipient = “example@gmail.com”

Subject-lines addressed to recipient = “Stock Update” + “Order Update”

Contact info = alias + phone number + First Name + Last Name

Dates = “Dates of all communication w/ recipient”

Frequency = “5” // # of times recipient has been emailed

Q-6

[Hemanshu, general data element for a product order]

Product	Name of Product
Code	Product Code (as it appears in database)
Upper Stock Limit	10,000
Lower Stock Limit	100
Supplier	List of Suppliers of Product
Continuous or Discrete	Discrete
Type	Numeric
Base of Derived	Base (product name is not derived from anything, ex. age is derived from date (which is base)
Comments	

[yash]

Provide a data element example for your system like slide 57.

```

=====
Name                New Template Body Text
Alias               Create New Template Body Text
Description         User Entered HTML text up to a length of 25 MB as imposed by Gmail
Mail Text+Attachment limit to be saved in Template Database as a new template.
Length             25,000,000
Input Format        X(25000000)
Output Format       X(25000000)
Default Value
Continuous/Discrete NA
Type               Character
Base or Derived     Base
Upper Limit        25,000,000
Lower Limit        1
Discrete           Text
Comments           The template must have at least one character entered in the template
entry editor window.
=====

```

Q-7

[Hemanshu, general data store for a product order]

Product ID	Name of Product
Upper Stock Limit	10,000
Lower Stock Limit	100
Supplier ID	List of Supplier IDs (Derived from Suppliers Table)
Record Size?	1 GB
Comments	

[yash]

Provide a Data Store Example for your system like slide

```

=====
ID                D1
Name              Template File
Alias             None

```

Description	Contains a record for each user created template	
File Type	Computer	
File Format	Database	
Record Size	25,000,020	(25,000,000 for the template text, 8 for unique record id, 8 usage counter, 4 last update datetime)
Maximum Records	1,000	
Average Records	128	
Percent Growth/Year	5%	
=====		

Submission 11

Q-1 How software management is different from other disciplines?

Software management is different from other disciplines due to the contributions of three factors.

- a. **Intangibility:** Software, unlike many other projects, is largely intangible. Most projects from other disciplines have some form of a tangible product or at least a working prototype which is referred to throughout the duration of the development. However, this is not the case for software— as a result, progress can be challenging to measure. It is naive to attempt to predict developer output, cannot extrapolate performance of one task and use it to predict future tasks. A highly-experienced developer may take a week for some tasks and only a few hours for others.
- b. **Short-lived:** Most software projects are developed once, and similar projects are rarely developed (since similar ones would simply be extended for short-term use). As a result, the challenges in developing new software solutions are often unforeseeable. Managers have a general idea of common issues, but the technical issues that may result vary between each project and managers do not have strategies prepared for the specific challenges that may arise.
- c. **Variable Processes:** Software projects vary significantly, and the firms that develop them are also dynamic systems. Although software processes exist for several categories of projects, the concept of following a process nonetheless follows a “cookie-cutter” mentality. At times, project managers within the same organization must utilize different methodologies.

Q-2 What are the important factors that influence project management?

As each software project has its own new and unique challenges, there are several factors that influence the management of these challenges. As explained above, project managers must reject the “cookie-cutter” mentality and use different strategies to mitigate the risk posed by the combination of factors below:

- Company size
- Software customers
- Software size
- Software type
- Organizational culture
- Software development processes

Q-3 What makes software risk management important?

Software risk management is important not only because of the novel, unforeseeable challenges discussed above, but also because of the high software failure rate attributed to poor requirements management. Therefore, project managers must learn to anticipate these risks, and by doing so they can manage the risks. First and foremost, the first place of software uncertainty originates from unclear software requirements. Non-technical customers and negligence of precise language can result in requirements which do not directly translate to effective analysis models. Furthermore, managers must be cognisant of the following variables:

- Timeline: when should the project be finished by?
- Budget
- Quality
- Task/Role Allocation (critical paths, based on talents, etc)
- Resources: Does your team have the appropriate resources required (software tools, hardware, licenses, etc)
- Daily progress review
- Conflict Resolution

Q-4 What are the important parameters for risk classification?

All risk classification can be grouped under two important parameters:

- The type of risk (technical, organizational, people-related, requirements-related and estimation risks)
- The entities/factors which are affected by the risk type chosen above

The three main entities which are affected by risks are; projects, products, and businesses. The list below encompasses the factors which are affected within each risk affectee:

1. Project risks affect schedule or resources;
2. Product risks affect the quality or performance of the software being developed;
3. Business risks affect the organisation developing or procuring the software.

Q-5 Write down briefly the risk management process.

The risk management process begins with “Risk Identification” which involves listing the potential risks your project contains. These can be risks to the organization, due to work culture, the timeline, conflict resolution, technical, etc. Next, “Risk Analysis” is performed, where risks are iteratively and continuously assigned values in a prioritized risk list (evaluated by severity/probability and the effects of the risk). Also, The consequences of each risk are also documented with great detail during this stage. Each identified risk is then addressed in a manner such that the risk can be avoided and contingency measures are developed (based on the priority of risks assigned in the “Risk Analysis” step. Finally, risks are assessed on an ongoing process throughout the development of the software solution. This happens during the “Risk Monitoring” stage and the monitored risks are used to iteratively and cyclically maintain and update the risk list (from the “Risk Analysis” stage) until project completion.

Q-6 What can be the elements of a checklist to identify common risks?

A checklist is often formed to identify common risks which a good project manager should review with their team. The following risks should be identified and described for your team’s specific project:

- Technology risks: Technology evolves rapidly, and large software projects inherit a risk of obsolescence. Managers of these projects must be cognizant that by the project end date, the solution may not be up-to-date or as timely. A project developed for Windows Vista, which takes 5+ years would prove to be less than ideal, as most users moved on to Windows 7 or back to Windows XP. Software dependent on third-party software or hardware is at greater risk than stand-alone solutions.
- Organizational risks: Bureaucracy within organizations exist and can vary dependent on both company size and company culture. Some teams are more at risk or change averse than others. Some companies have open-door policies whereas others use a more formalized communication strategy, and each poses its own risk in increasing project length.

- People risks: Recruitment of team members vary on a per project basis, and thus the skillset of each member is also a variable which is difficult to quantify. Some teams may suffer from redundant skillsets and the employees may not have the complementing talents needed to complete all aspects of the project; which leads to conflict. Non-normative events such as illness and compassionate leave are also usually unexpected, but can have drastic effects on the timeline, if contingencies are not developed in advance.
- Requirements risks: Changes to requirements that involve major design rework and analysis redesign may require all construction activities to be restarted. This risk is compounded and more severe if the customers are non-technical or if a core-requirement was ambiguous in implementation during the requirements analysis stage.
- Estimation risks: Just as ambiguous language in the “software requirements” document may pose requirements' risks, they also pose risks related to both the timeline of the project and the construction activity phases. As mentioned prior, the same developer may take various lengths of time for seemingly similar tasks, and every software solution is different from the ones developed prior. This makes it easy to underestimate seemingly similar software tasks, in not only time required for completion, but also the size of software and the rate of defect repair.

Q-7 Prepare a list of risks for your project on the following pattern.

Risk	Probability	Effects
1. Browser provider may stop the support for the add-on functionality	Moderate	Users will lose this key feature required by them
2. Browser provider may develop equivalent add-on to make it part of the native browser product.	Low	Users may be confused between the two and may need

		clarification on pros and cons of each.
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Q-8 Consider each risk and develop a strategy to manage that risk.

1. Advertise the features of the add-on to as many users as possible within the organisation as well as outside the organisation, so that the browser provider can track the usage of the add-on to continue to support this important feature.
2. Make the developed product feature rich and publish early to the market so that we have the advantage of entering the market first. By having more features useful to the users, the functions can be patented.

Q-9 How risk monitoring can be helpful for risk management?

As part of risk monitoring when we assess each identified risks regularly to decide whether or not it is becoming less or more probable, it allows us to focus on other important tasks of delivering the software product on time and within budget if the risk is less probable and it allows us to find ways to either avoid, minimise or handle them if they are more probable.

When we assess whether the effects of the risks have changed or not, we are able to warn the management of the adverse effects and take necessary action to prevent major impact on client organisations operations or the financial position if the risks have increased, or on the other hand if the risks have reduced, then it allows to make more bold investment in the product to achieve higher than expected returns from the product development and deployment.

When we discuss each key risk at the management progress meetings, it would allow us to make management aware of the potential of the increase or decrease of specific risks to make sound decisions on time to continue with the product development or to stop the development to prevent further financial loss.

Q-10 What are People management factors? Describe all of them.

Following are people management factors:

Consistency: Team members should all be treated in a comparable way without favourites or discrimination.

Respect: Different team members have different skills and these differences should be respected.

Inclusion: Involve all team members and make sure that people's views are considered.

Honesty: You should always be honest about what is going well and what is going badly in a project.

Q-11 What are important factors required to be considered for assembling a team, describe each of them and their impact.

Following are the important factors required to be considered for assembling a team:

- Group Composition
- Group Organisation
- Informal Groups
- Group Communications
- Work within the following constraints:
 1. May not be possible to appoint the ideal people to work on a project
 2. Project budget may not allow for the use of highly-paid staff;
 3. Staff with the appropriate experience may not be available;
 4. An organisation may wish to develop employee skills on a software project.

Group composition: Group composed of members who share the same motivation can be problematic

Task-oriented - everyone wants to do their own thing;

Self-oriented - everyone wants to be the boss;

Interaction-oriented - too much chatting, not enough work.

An effective group has a balance of all types.

This can be difficult to achieve software engineers are often task-oriented.

Interaction-oriented people are very important as they can detect and defuse tensions that arise.

Group Organisation:

The way that a group is organized affects the decisions that are made by that group, the ways that information is exchanged and the interactions between the development group and external project stakeholders.

Key questions include:

1. Should the project manager be the technical leader of the group?
2. Who will be involved in making critical technical decisions, and how will these be made?
3. How will interactions with external stakeholders and senior company management be handled?
4. How can groups integrate people who are not co-located?
5. How can knowledge be shared across the group?

Small software engineering groups are usually organised informally without a rigid structure.

For large projects, there may be a hierarchical structure where different groups are responsible for different sub-projects.

Agile development is always based around an informal group on the principle that formal structure inhibits information exchange

Informal Groups:

- The group acts as a whole and comes to a consensus on decisions affecting the system.
- The group leader serves as the external interface of the group but does not allocate specific work items.
- Rather, work is discussed by the group as a whole and tasks are allocated according to ability and experience.
- This approach is successful for groups where all members are experienced and competent.

Group Communications:

Good communications are essential for effective group working.

Information must be exchanged on the status of work, design decisions and changes to previous decisions.

Good communications also strengthens group cohesion as it promotes understanding.

Group size

The larger the group, the harder it is for people to communicate with other group members.

Group structure

Communication is better in informally structured groups than in hierarchically structured groups.

Group composition

Communication is better when there are different personality types in a group and when groups are mixed rather than single sex.

The physical work environment

Good workplace organisation can help encourage communications.

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Gmail Template Engine: User Manual

Product Name

Gmail Template Engine

Product Front-End UI

Compose UI

The screenshot displays the Gmail 'Compose' interface. On the left, an email draft is visible with the subject 'our spreadsheet', recipient 'Alex Ladd <ccps731dev@gmail.com>', and body text 'works?'. Below the draft is a virus-free notification from Avast. At the bottom of the draft area is a 'Send' button and a row of icons for text formatting, attachments, emojis, and more. On the right, the 'Email Templator' sidebar is open, showing a list of templates. 'Template 12' is selected, indicated by a green circle. Below the list, the preview of the selected template is shown, containing a project update message with placeholder text for project name, supplier, and dates. The template ends with 'Thanks,'.

our spreadsheet

Alex Ladd <ccps731dev@gmail.com>
to me ▾
works?

Virus-free. www.avast.com

← ▾ Alex Ladd (ccps731dev@gmail.com)

Send ▾

Email Templator ⋮ ✕

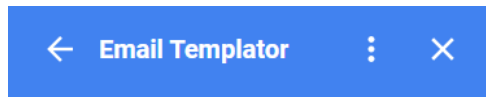
- ☐ Template 9
- ☐ Template 10
- ☐ Template 11
- ☒ Template 12
- ☐ Yashs template one
- ☐ Alex's Template
- ☐ Test Multi-line Input
- ☐ Yashs template four

Project _____ Update

Supplier _____ has stated their delivery will be delayed for another _____ days. As a result, the materials will now not be available until __ - __ - __. I ask all team leads to report to my office if supplier _____'s delay will impede their department functions.

Thanks,

Message UI



Create a New Template

[CREATE TEMPLATE](#) [RESET](#)

Template Name:

Template Content:

License/Terms of Use

The Gmail Template Engine does not guarantee access of stored templates. All stored templates are stored using Google's Sheets service, and the service utilized is each user's personal Sheets application. Gmail Template Engine is not liable for any lost or stolen data. Google's middleware and intermediate storage solutions are used for the making of this app and Gmail Template Engine is simply an extension of those Google services. Gmail Template engine is currently issued on a GNU/GPL license.

Authorizations Requested

Gmail Template Engine does not store or collect any email information from users. All template information is stored on each individual's personal user accounts, and Gmail Template Engine does not store any information from users. No personally identifiable information is logged;

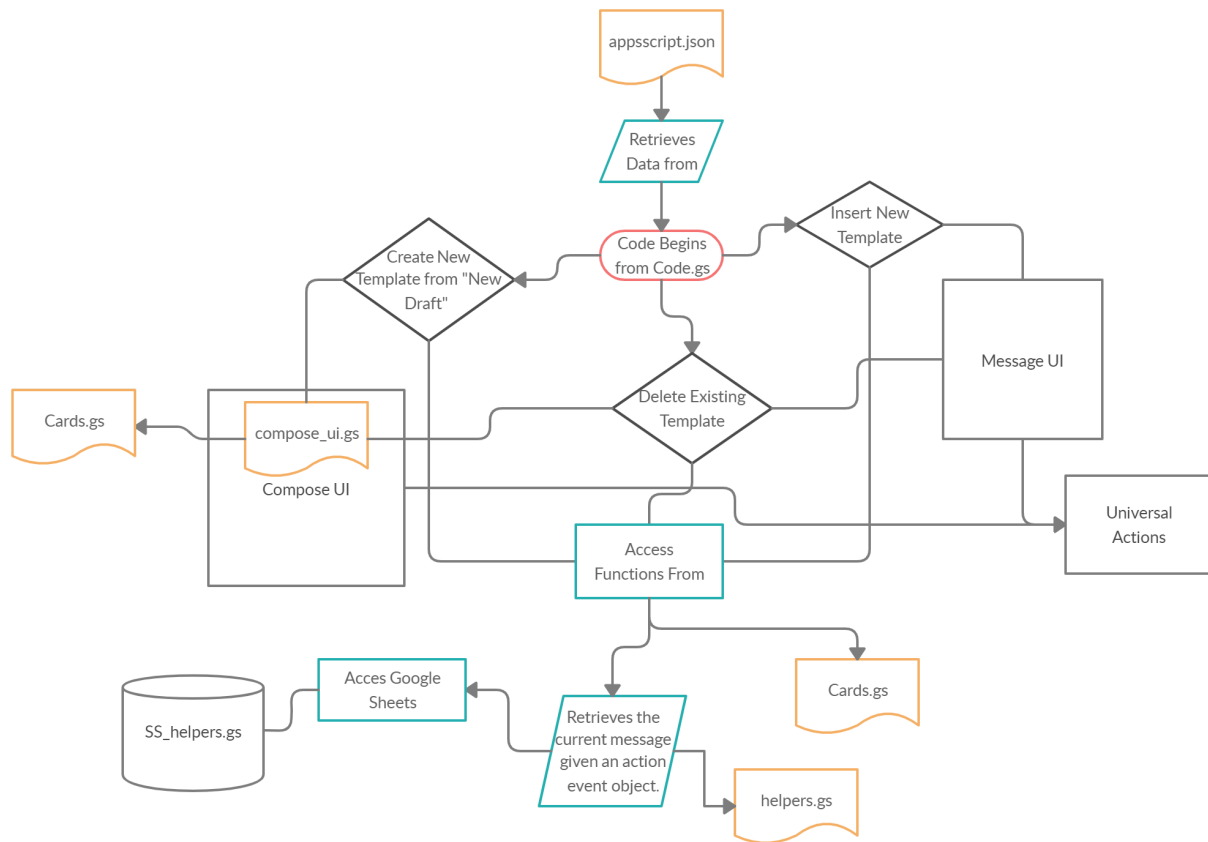
each user's data stays within the confines of their add-on's allocated Google Sheet. With that said, the following authorizations must be accepted in order to utilize Gmail Template Engine:

View your email messages when the add-on is running	https://www.googleapis.com/auth/gmail.addons.current.message.readonly
View your country, language, and timezone	https://www.googleapis.com/auth/script.locale
Manage drafts and send emails when you interact with the add-on	https://www.googleapis.com/auth/gmail.addons.current.action.compose
See, edit, create, and delete your spreadsheets in Google Drive	https://www.googleapis.com/auth/spreadsheets
View your email message metadata such as labels and headers, but not the email body	https://www.googleapis.com/auth/gmail.metadata
Run as a Gmail add-on	https://www.googleapis.com/auth/gmail.addons.execute
View and modify but not delete your email	https://www.googleapis.com/auth/gmail.modify
View your email message metadata when the add-on is running	https://www.googleapis.com/auth/gmail.addons.current.message.metadata

Logical Flow of Source Code

Below is a logical flow of the interconnected activities which take place during the execution of the Gmail Template Engine add-on. Although the flow chart encompasses all files and major entities, it is only a conceptual mapping and does not necessarily represent the “physical” implementation of the source code. The add-on consists of two application windows (pictured above, in the “Product Front-End UI” section), which stem from the two states that template creation exists in: the “compose UI” and the “message UI.” Three decisions are possible in the add-on, template creation, deletion and insertion into the repository. All files access both the helpers.gs (a function of utility helper functions, as the name implies) and a version of cards.gs (the “card” layout). “SS_helpers.gs” is a file which interacts with the database (Google Sheets,

this choice of database is explained later in the report) and then is accessed by both templating states.



Technical specifications

All templates are limited to the specifications of which limit all Gmail messages. Templates are unable to store attachments for privacy and security reasons. Template storage is also limited to each user's personal storage available to their Gmail accounts. In order to ensure zero-latency in template retrieval, Gmail Template Engine uses each individual's personal/commercial Google Sheets. Google Sheets storage capacity is limited under the the overall space allotment Google provides each of their users.

Why Google Sheets?

Google Sheets is the medium of choice for the corpus because Google's services work best when they are integrated together. Google stores all of its users' data on their own servers and allows its account users access to that data by interacting with its services. Developers can also access data of their clients, as long as their clients agree to the requested permissions. As a result, Google has created several functions, which one can also view as "black-boxes", which developers can use specifically to store account data into other Google services. Ideally one would implement a database for such programs (where the corpus and inverted index can be hosted). However, this solution fails for two reasons. First, Google's data is stored online, so it is impractical to attach a local database for every download of an add-on. Second, developers cannot use Google's cloud database options either, and connecting to individual accounts is not easy nor secure. Therefore, Google Sheets provides a lightweight storage solution, which can be manipulated by Google's own programming language and is available to all users of the add-on by default (since every Gmail account comes packaged with access to Google Sheets).

The spreadsheets will pull data from emails (performed in my other collaborative effort), and this search tool would allow me to fix a missing link in that project. Potentially, the add-on may allow clients to upload properly formatted spreadsheets (meeting our requirements), which would allow instant, mass storage of a pre-existing template repository. The goal is to create a practical program, which can search for both "Subject" line, and email body text, as these fields grow with daily incoming emails.

Glossary

Google Sheets: A spreadsheet web application which is provided by Google, for all Gmail account holders. Gmail Template Engine does not issue Google Sheets, rather, they create a Google Sheet instance for each installation of the add-on.

Google Apps Scripts: A platform created by Google for developers to create products which extend Google services.

Template: An email message which has been stored for future-use. A template consists of three parts, the receiver, subject line, and the 'body' of the email message.

Database: All references made to a database denotes the strict use of the individual's personal/commercial Google Sheets, found under the list of applications provided to users of Google accounts (associated with every Gmail, free of charge).

Trigger: In the context of the operation of the Gmail Template Engine, a trigger refers to a database trigger (as explained above, database refers to Google Sheets). Any “trigger” implies that the user has clicked or interacted with the Gmail Template Engine in a manner which results in some data manipulation of the database.

Value Proposition

Objective

Many operations’ (marketing, sales, production, supply chain, IT, customer service, HR etc.) staff often sends confidential messages– in a sense, they are for intended recipients only. However, many of these are common in content, at least through the perspective of users sending them on a daily basis. Our tool seeks to not automate the task of emailing, but rather empower the user to create their own email template engine.

We intend to create a customizable Gmail add-on which allows email templating and personalized message delivery. The email add-on will host a light-weight database (the size of a small table) which will be populated based on the custom user profiles. The user will have the freedom to design prefilled messages for as many scenarios as they require, with one-click execution to automatically fill the selected/current email message with the desired template.

The human aspect of the interface is critical and relies on the user’s domain knowledge to customize their profile with their unique needs–rather than simply relying on generic profiles force fed to the consumer. The user is given the power to save their own custom messages for specific scenarios without any need to write a line of code. Furthermore, the user has all of these messages available from the touch of a button. A simple click will allow the user to automatically fill their selected message with the desired custom message– so that it can be sent or modified as the user sees fit.

We hope to fulfill the following requirements for consumers, by providing:

- A custom templating feature
- Easy to click buttons
- History of templates available
- No required storage management
- Autofill draft message

Target Customers

Working professionals in dire need of time in their busy lifestyles. This tool can be used by anyone who uses email as a significant communication channel in their daily life. Our focus will cater on those who send frequent repetitive emails to individuals, emails which cannot be automated as the recipients and content differs slightly each time, but nonetheless is repetitive for the user.

Google Account Holders Only: The Gmail template engine is intended for use only with Gmail accounts. It is not limited to personal or commercial accounts, and can be freely used with any legitimate Gmail account.

Product Release Schedule

1st Iteration

- Allow users to create a profile and save custom email templates for various scenarios
- Extract/read subject-line of current/selected email
- Use pre-populated database to auto-generate emails

2nd Iteration

- Move most frequently used templates under “recent section”
- Add search engine/text retrieval system

→ Product is ready to deploy

3rd Iteration

- List Google Calendar events for possible automated actions
- Log user behaviour (based on previous templates used, suggest most frequently used templates)
- Bonus: Use machine learning algorithms or available Google intelligence to suggest templates based on past responses (regardless of frequency of use)

Technical Descriptions

Below is a list of all of the features that Gmail Template Engine provides, along with input, output and source descriptions.

Gmail Add-on Feature 1

Function: Display "Choose a Reply Template" selection Pop-up

Description: Display the "Choose a Reply Template" selection Pop-up when the user selects a single email from the inbox to reply to it and chooses this option from the Add-on menu.

Inputs: None

Source: User clicks on an email in their Gmail inbox to reply to the sender, then clicks



on the Add-on icon.

Outputs: "Choose a Reply Template" window displayed to user's desktop screen on the right side

Destination: Template Database Open function.

Gmail Add-on Feature 2

Function: Open Template Database

Description: Open Template Database (Google Spreadsheet) that is expected to be available in the Add-on location.

If the database is not available, then issue an error to the user and return control back to the user.

At this point, the Template Database is expected to be available.

Populate the "Email Templator" pop-up window with a radio button for user to select it.

Inputs: Template Database (Google Spreadsheet) with a sheet named 'Email Templates'

Source: User click to the "Choose a Reply Template"

Outputs: “Email Templator” pop-up window populated with all templates from the database.

Destination: Select Template from available templates.

Gmail Add-on Feature 3

Function: Select Template from available templates

Description: When user clicks a particular template and clicks the “Insert Template” button, retrieve the template email text either from memory array that corresponds to the selected template.

Once the Template email text is added to the email reply, the email being replied gets saved in the “Drafts” folder for user. The user would then select it from the draft to update any additional detail to fill in the blanks in the template email text.

Inputs: User selection of a template from the top section of “Choose a Reply Template” window.

Template Database (Google Spreadsheet)

Array of template row# to template text cross-reference list in memory.

Source: Open Template Database (Google Spreadsheet)

Outputs: User’s reply email populated with the template email text selected by the user and saved in the “Drafts” folder.

Destination: User’s reply email.

Gmail Add-on Feature 4

Function: Exit Template Selection window

Description: When user clicks the “X” button on the top right hand corner of the “Choose a Reply Template” window, close the “Choose a Reply Template” window and return to the User’s inbox window.

Inputs: User action to close the “Choose a Reply Template” window.

Source: User selection of an email in his/her inbox.

Outputs: Closure of "Choose a Reply Template" window.

Destination: User's inbox.

Gmail Add-on Feature 5 (To be deployed during Iteration #2)

Function: Parse for search keywords in selected email

Description: Parse the words from the subject line of selected email.

if the subject line is empty, then parse the words from the first paragraph of the email body.

Build unique list of keywords to search in the Template Database's keyword field.

If the unique list of parsed keywords is empty, then (1) Display the information message, (2) Do not attempt to search the Template Database, (3) Disable "Choose a Reply Template" button, so that user is only able to exit out of the Template Selection window.

Inputs: Selected email (subject line and first paragraph of the email body text),

Template Database (Google Spreadsheet)

Source: Open Template Database

Outputs: Unique List of search keywords

Destination: Search Template Database for keyword match

Gmail Add-on Feature 6 (To be deployed during Iteration #2)

Function: Search Template Database for keyword match

Description: Search Template Database in two ways.

(1) Concatenate multiple parsed keywords found from email to create a composite keyword to be searched in the Template Database's keyword field

Find templates with composite keyword

(2) Use the individual parsed keywords found from the email to search directly in the Template Database's keyword field.

Find templates with individual parsed keywords.

Populate all template messages found into the Middle section of the Template Selection window labeled “Matched Templates Available for Selection”

Also build and maintain the array of

(a) composite keyword or individual parsed keyword along with corresponding

(b) template email text in memory or the sequence number of the records populated in the top section of “Select a Template to Insert into an Email” window. This field (b) will be used to either populate the user’s reply email or to do the lookup in Template Database to get the template email text for the purpose of populating the user’s reply email.

Get the count of matched templates.

If the count is > 0, then enable the “Insert Template into Email Reply” button in the “Select a Template to Insert into an Email” window for user to select one or more template.

Inputs: Unique List of search keywords,
Template Database (Google Spreadsheet)

Source: Parse for search keywords in selected email

Outputs: Populated Template Selection window for top section.
Array of keyword to template cross-reference list in memory.

Destination: Select Template from matched templates

Gmail Add-on Feature 7

Function: Delete an existing Template

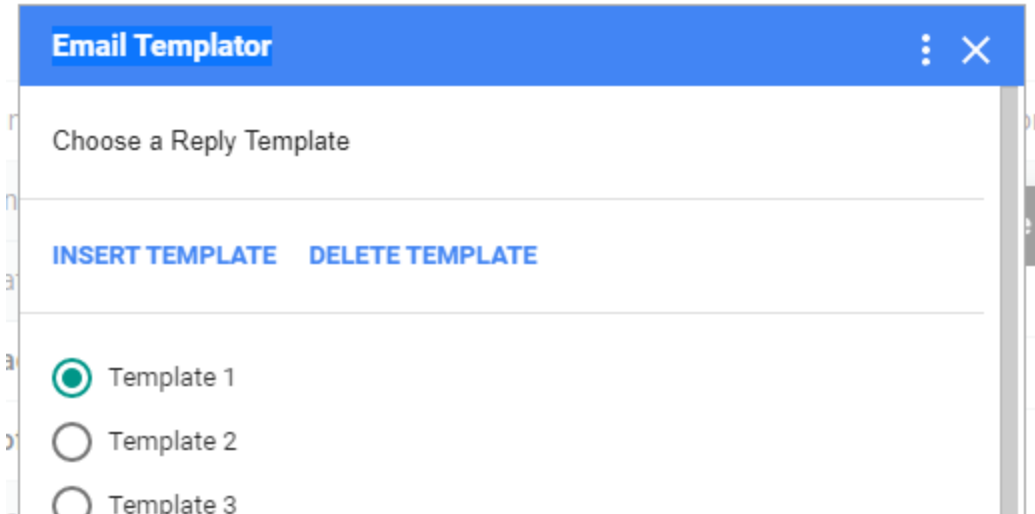
Description: When user selects to compose a new email draft, the add-on provides Delete Template function to delete an existing template.

Inputs: User action to compose a new email.

Source: User’s inbox

Outputs: Updated Email Templator window with the selected template removed from the list and the Template Database (Google Spreadsheet) updated.

Destination: User stays in the Email Templator window.



Gmail Add-on Feature 8

Function: Create a New Template

Description: When user selects an email from the inbox, the add-on opens the "Email Templator" pop-up window with two options: (1) Choose a Reply Template and (2) Create a New Template. Upon click of the "Create a New Template", the "Create a New Template" pop-up opens for user to enter the template name and the template text. Once the template name and template text is entered, the user can click the "CREATE TEMPLATE" button to save the new template into the Template Database.

Email Templator

Choose a Reply Template

Create a New Template

←

Email Templator

Create a New Template

CREATE TEMPLATE

RESET

Template Name:

Template Content:

← Email Templator
⋮
✕

Create a New Template

CREATE TEMPLATE
RESET

Template Name:
Daily Scrum notes

Template Content:
Attendees: Yash, Alex, Hemanshu
Notes:
Yash: What did you accomplish yesterday:
What you plan to do today:
Any roadblock to be removed:
Alex: What did you accomplish yesterday:
What you plan to do today:
Any roadblock to be removed:
Hemanshu: What did you accomplish
yesterday:
What you plan to do today:
Any roadblock to be removed:

Inputs: User action to select an existing email, then the entered form with new template information to be saved in the Template Database.

Source: User's inbox

Outputs: Updated Template Database (Google Spreadsheet) with one additional row.

←

Email Templator

⋮

×

Template 5

Template 6

Template 7

Template 9

Template 10

Template 11

Template 12

Yashs template one

Alex's Template

Test Multi-line Input

Yashs template four

Daily Scrum notes

Attendees: Yash, Alex, Hemanshu

Notes:

Yash: What did you accomplish yesterday:
What you plan to do today:
Any roadblock to be removed:

Alex: What did you accomplish yesterday:
What you plan to do today:
Any roadblock to be removed:

Hemanshu: What did you accomplish yesterday:
What you plan to do today:
Any roadblock to be removed:

Destination: User stays in the Create a New Template window.

Installation instructions

Upon completion, the application can be installed in the same manner as any Gmail add-on.

To download our add-on, or any add-on:

1. Simply login to your Gmail account
2. Click the side-panel
3. Click the '+' sign for "Get Add-ons"
 - a. Search for "Gmail Template Add-on"
4. If the side-panel is not available, Google Chrome users can alternative use the Chrome Web Store to download the add-on using the following procedure:
 - a. Access the menu button on your chrome browser
 - b. Click "Extensions"
 - c. Search for "Gmail Template Add-on"
 - d. Click download

Contact Info

Have a question, a compliment, or a business inquiry? Please leave your feedback with the developers at ccps731dev@gmail.com. We look forward to hearing from you!